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CURING IMPAIRED CONTENT UTILIZING A KNOWLEDGE DATABASE OF ENTIGENS

Abstract

A method performed by a computing device includes obtaining impaired content for a topic that includes a series of words, and determining a set of identigens for each word of the series of words to produce sets of identigens. The method further includes interpreting, using identigen pairing rules of a knowledge database, the sets of identigens to determine a most likely meaning interpretation of the impaired content and produce an interim entigen group. The method further includes recovering an improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on a knowledge defect of the interim entigen group caused by an impairment of the impaired content.

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Background/Summary

CROSS REFERENCE TO RELATED PATENTS [0001] The present U.S. Utility Patent application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 17/842,095, entitled “CURING IMPAIRED CONTENT UTILIZING A KNOWLEDGE DATABASE OF ENTIGENS,” filed Jun. 16, 2022, issuing on May 27, 2025 as U.S. Pat. No. 12,314,237, which claims priority pursuant to 35 U.S.C. § 120 as a continuation-in-part of U.S. Utility application Ser. No. 16/737,123, entitled “CONVERTING CONTENT FROM A FIRST TO A SECOND APTITUDE LEVEL,” filed Jan. 8, 2020, issued Jul. 12, 2022 as U.S. Pat. No. 11,386,130, which claims priority pursuant to 35 U.S.C. § 119 (e) to U.S. Provisional Application No. 62/797,454, entitled “TRANSLATING CONTENT FROM A FIRST TO A SECOND APTITUDE LEVEL,” filed Jan. 28, 2019, expired, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility Patent Application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT—NOT APPLICABLE
INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC—NOT APPLICABLE
BACKGROUND OF THE INVENTION
Technical Field of the Invention
[0002] This invention relates generally to computing systems and more particularly to generating data representations of data and analyzing the data utilizing the data representations.
Description of Related Art
[0003] It is known that data is stored in information systems, such as files containing text. It is often difficult to produce useful information from this stored data due to many factors. The factors include the volume of available data, accuracy of the data, and variances in how text is interpreted to express knowledge. For example, many languages and regional dialects utilize the same or similar words to represent different concepts.
[0004] Computers are known to utilize pattern recognition techniques and apply statistical reasoning to process text to express an interpretation in an attempt to overcome ambiguities inherent in words. One pattern recognition technique includes matching a word pattern of a query to a word pattern of the stored data to find an explicit textual answer. Another pattern recognition

technique classifies words into major grammatical types such as functional words, nouns, adjectives, verbs and adverbs. Grammar based techniques then utilize these grammatical types to study how words should be distributed within a string of words to form a properly constructed grammatical sentence where each word is forced to support a grammatical operation without necessarily identifying what the word is actually trying to describe.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0005] FIG. **1** is a schematic block diagram of an embodiment of a computing system in accordance with the present invention;

[0006] FIG. **2** is a schematic block diagram of an embodiment of various servers of a computing system in accordance with the present invention;

[0007] FIG. **3** is a schematic block diagram of an embodiment of various devices of a computing system in accordance with the present invention;

[0008] FIGS. **4**A and **4**B are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;

[0009] FIG. **4**C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system in accordance with the present invention;

[0010] FIG. **5**A is a schematic block diagram of an embodiment of a collections module of a computing system in accordance with the present invention;

[0011] FIG. **5**B is a logic diagram of an embodiment of a method for obtaining content within a computing system in accordance with the present invention;

[0012] FIG. **5**C is a schematic block diagram of an embodiment of a query module of a computing system in accordance with the present invention;

[0013] FIG. **5**D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system in accordance with the present invention;

[0014] FIG. **5**E is a schematic block diagram of an embodiment of an identigen entigen intelligence (IEI) module of a computing system in accordance with the present invention;

[0015] FIG. **5**F is a logic diagram of an embodiment of a method for analyzing content within a computing system in accordance with the present invention;

[0016] FIG. **6**A is a schematic block diagram of an embodiment of an element identification module and an interpretation module of a computing system in accordance with the present invention;

[0017] FIG. **6**B is a logic diagram of an embodiment of a method for interpreting information within a computing system in accordance with the present invention;

[0018] FIG. **6**C is a schematic block diagram of an embodiment of an answer resolution module of a computing system in accordance with the present invention;

[0019] FIG. **6**D is a logic diagram of an embodiment of a method for producing an answer within a computing system in accordance with the present invention;

[0020] FIG. **7**A is an information flow diagram for interpreting information within a computing system in accordance with the present invention;

[0021] FIG. **7**B is a relationship block diagram illustrating an embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;

[0022] FIG. **7**C is a diagram of an embodiment of a synonym words table within a computing system in accordance with the present invention;

[0023] FIG. **7**D is a diagram of an embodiment of a polysemous words table within a computing system in accordance with the present invention;

[0024] FIG. **7**E is a diagram of an embodiment of transforming words into groupings within a computing system in accordance with the present invention;
[0025] FIG. **8**A is a data flow diagram for accumulating knowledge within a computing system in accordance with the present invention;
[0026] FIG. **8**B is a diagram of an embodiment of a groupings table within a computing system in accordance with the present invention;
[0027] FIG. **8**C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system in accordance with the present invention;
[0028] FIG. **8**D is a data flow diagram for answering questions utilizing interference within a computing system in accordance with the present invention;
[0029] FIG. **8**E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system in accordance with the present invention;
[0030] FIGS. **8**F and **8**G are schematic block diagrams of another embodiment of a computing system in accordance with the present invention;
[0031] FIG. **8**H is a logic diagram of an embodiment of a method for processing content to produce knowledge within a computing system in accordance with the present invention;
[0032] FIGS. **8**J and **8**K are schematic block diagrams another embodiment of a computing system in accordance with the present invention;
[0033] FIG. **8**L is a logic diagram of an embodiment of a method for generating a query response to a query within a computing system in accordance with the present invention;
[0034] FIGS. **9**A and **9**B are schematic block diagrams of another embodiment of a computing system illustrating a method for converting content of a first aptitude level to converted content of a second aptitude level within the computing system in accordance with the present invention.
[0035] FIG. **10**A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;
[0036] FIG. **10**B is a data flow diagram of an embodiment of formalizing content within a computing system in accordance with the present invention;
[0037] FIG. **10**C is a logic diagram of an embodiment of a method for formalizing content within a computing system in accordance with the present invention;
[0038] FIG. **11**A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;
[0039] FIG. **11**B is structure diagram of an embodiment of a knowledge database within a computing system in accordance with the present invention;
[0040] FIG. **11**C is a logic diagram of an embodiment of a method for accessing a knowledge database within a computing system in accordance with the present invention;
[0041] FIG. **12**A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;
[0042] FIG. **12**B is a structure diagram of another embodiment of a knowledge database within a computing system in accordance with the present invention;
[0043] FIG. **12**C is a logic diagram of another embodiment of a method for accessing a knowledge database within a computing system in accordance with the present invention;
[0044] FIG. **13**A is a schematic block diagram of another embodiment of a computing system in accordance with the present invention;
[0045] FIG. **13**B is a data flow diagram of an embodiment for correcting impaired content within a computing system in accordance with the present invention;
[0046] FIG. **13**C is a logic diagram of an embodiment of a method for correcting impaired content within a computing system in accordance with the present invention; and
[0047] FIGS. **13**D-**13**E are schematic block diagrams of another embodiment of a computing system illustrating another method for correcting impaired content within the computing system in

accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0048] FIG. **1** is a schematic block diagram of an embodiment of a computing system **10** that includes a plurality of user devices **12-1** through **12**-N, a plurality of wireless user devices **14-1** through **14**-N, a plurality of content sources **16-1** through **16**-N, a plurality of transactional servers **18-1** through **18**-N, a plurality of artificial intelligence (AI) servers **20-1** through **20**-N, and a core network **24**. The core network **24** includes at least one of the Internet, a public radio access network (RAN), and any private network. Hereafter, the computing system **10** may be interchangeably referred to as a data network, a data communication network, a system, a communication system, and a data communication system. Hereafter, the user device and the wireless user device may be interchangeably referred to as user devices, and each of the transactional servers and the AI servers may be interchangeably referred to as servers.

[0049] Each user device, wireless user device, transactional server, and AI server includes a computing device that includes a computing core. In general, a computing device is any electronic device that can communicate data, process data, and/or store data. A further generality of a computing device is that it includes one or more of a central processing unit (CPU), a memory system, a sensor (e.g., internal or external), user input/output interfaces, peripheral device interfaces, communication elements, and an interconnecting bus structure.

[0050] As further specific examples, each of the computing devices may be a portable computing device and/or a fixed computing device. A portable computing device may be an embedded controller, a smart sensor, a smart pill, a social networking device, a gaming device, a cell phone, a smart phone, a robot, a personal digital assistant, a digital music player, a digital video player, a laptop computer, a handheld computer, a tablet, a video game controller, an engine controller, a vehicular controller, an aircraft controller, a maritime vessel controller, and/or any other portable device that includes a computing core. A fixed computing device may be security camera, a sensor device, a household appliance, a machine, a robot, an embedded controller, a personal computer (PC), a computer server, a cable set-top box, a satellite receiver, a television set, a printer, a fax machine, home entertainment equipment, a camera controller, a video game console, a critical infrastructure controller, and/or any type of home or office computing equipment that includes a computing core. An embodiment of the various servers is discussed in greater detail with reference to FIG. **2**. An embodiment of the various devices is discussed in greater detail with reference to FIG. **3**.

[0051] Each of the content sources **16-1** through **16**-N includes any source of content, where the content includes one or more of data files, a data stream, a tech stream, a text file, an audio stream, an audio file, a video stream, a video file, etc. Examples of the content sources include a weather service, a multi-language online dictionary, a fact server, a big data storage system, the Internet, social media systems, an email server, a news server, a schedule server, a traffic monitor, a security camera system, audio monitoring equipment, an information server, a service provider, a data aggregator, and airline traffic server, a shipping and logistics server, a banking server, a financial transaction server, etc. Alternatively, or in addition to, one or more of the various user devices may provide content. For example, a wireless user device may provide content (e.g., issued as a content message) when the wireless user device is able to capture data (e.g., text input, sensor input, etc.).

[0052] Generally, an embodiment of this invention presents solutions where the computing system supports the generation and utilization of knowledge extracted from content. For example, the AI servers **20-1** through **20**-N ingest content from the content sources **16-1** through **16**-N by receiving, via the core network **24** content messages **28-1** through **28**-N as AI messages **32-1** through **32**-N, extract the knowledge from the ingested content, and interact with the various user devices to utilize the extracted knowledge by facilitating the issuing, via the core network **24**, user messages **22-1** through **22**-N to the user devices **12-1** through **12**-N and wireless signals **26-1** through **26**-N to the wireless user devices **14-1** through **14**-N.

[0053] Each content message **28-1** through **28**-N includes a content request (e.g., requesting content related to a topic, content type, content timing, one or more domains, etc.) or a content response, where the content response includes real-time or static content such as one or more of dictionary information, facts, non-facts, weather information, sensor data, news information, blog information, social media content, user daily activity schedules, traffic conditions, community event schedules, school schedules, user schedules airline records, shipping records, logistics records, banking records, census information, global financial history information, etc. Each AI message **32-1** through **32**-N includes one or more of content messages, user messages (e.g., a query request, a query response that includes an answer to a query request), and transaction messages (e.g., transaction information, requests and responses related to transactions). Each user message **22-1** through **22**-N includes one or more of a query request, a query response, a trigger request, a trigger response, a content collection, control information, software information, configuration information, security information, routing information, addressing information, presence information, analytics information, protocol information, all types of media, sensor data, statistical data, user data, error messages, etc.
[0054] When utilizing a wireless signal capability of the core network **24**, each of the wireless user devices **14-1** through **14**-N encodes/decodes data and/or information messages (e.g., user messages such as user messages **22-1** through **22**-N) in accordance with one or more wireless standards for local wireless data signals (e.g., Wi-Fi, Bluetooth, ZigBee) and/or for wide area wireless data signals (e.g., 2G, 3G, 4G, 5G, satellite, point-to-point, etc.) to produce wireless signals **26-1** through **26**-N. Having encoded/decoded the data and/or information messages, the wireless user devices **14-1** through **14**-N and/receive the wireless signals to/from the wireless capability of the core network **24**.
[0055] As another example of the generation and utilization of knowledge, the transactional servers **18-1** through **18**-N communicate, via the core network **24**, transaction messages **30-1** through **30**-N as further AI messages **32-1** through **32**-N to facilitate ingesting of transactional type content (e.g., real-time crypto currency transaction information) and to facilitate handling of utilization of the knowledge by one or more of the transactional servers (e.g., for a transactional function) in addition to the utilization of the knowledge by the various user devices. Each transaction message **30-1** through **30**-N includes one or more of a query request, a query response, a trigger request, a trigger response, a content message, and transactional information, where the transactional information may include one or more of consumer purchasing history, crypto currency ledgers, stock market trade information, other investment transaction information, etc.
[0056] In another specific example of operation of the generation and utilization of knowledge extracted from the content, the user device **12-1** issues a user message **22-1** to the AI server **20-1**, where the user message **22-1** includes a query request and where the query request includes a question related to a first domain of knowledge. The issuing includes generating the user message **22-1** based on the query request (e.g., the question), selecting the AI server **20-1** based on the first domain of knowledge, and sending, via the core network **24**, the user message **22-1** as a further AI message **32-1** to the AI server **20-1**. Having received the AI message **32-1**, the AI server **20-1** analyzes the question within the first domain, generates further knowledge, generates a preliminary answer, generates a quality level indicator of the preliminary answer, and determines to gather further content when the quality level indicator is below a minimum quality threshold level.
[0057] When gathering the further content, the AI server **20-1** issues, via the core network **24**, a still further AI message **32-1** as a further content message **28-1** to the content source **16-1**, where the content message **28-1** includes a content request for more content associated with the first domain of knowledge and in particular the question. Alternatively, or in addition to, the AI server **20-1** issues the content request to another AI server to facilitate a response within a domain associated with the other AI server. Further alternatively, or in addition to, the AI server **20-1** issues the content request to one or more of the various user devices to facilitate a response from a subject

matter expert.
[0058] Having received the content message **28-1**, the contents or **16-1** issues, via the core network **24**, a still further content message **28-1** to the AI server **20-1** as a yet further AI message **32-1**, where the still further content message **28-1** includes requested content. The AI server **20-1** processes the received content to generate further knowledge. Having generated the further knowledge, the AI server **20-1** re-analyzes the question, generates still further knowledge, generates another preliminary answer, generates another quality level indicator of the other preliminary answer, and determines to issue a query response to the user device **12-1** when the quality level indicator is above the minimum quality threshold level. When issuing the query response, the AI server **20-1** generates an AI message **32-1** that includes another user message **22-1**, where the other user message **22-1** includes the other preliminary answer as a query response including the answer to the question. Having generated the AI message **32-1**, the AI server **20-1** sends, via the core network **24**, the AI message **32-1** as the user message **22-1** to the user device **12-1** thus providing the answer to the original question of the query request.
[0059] FIG. **2** is a schematic block diagram of an embodiment of the AI servers **20-1** through **20**-N and the transactional servers **18-1** through **18**-N of the computing system **10** of FIG. **1**. The servers include a computing core **52**, one or more visual output devices **74** (e.g., video graphics display, touchscreen, LED, etc.), one or more user input devices **76** (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.), one or more audio output devices **78** (e.g., speaker(s), headphone jack, a motor, etc.), and one or more visual input devices **80** (e.g., a still image camera, a video camera, photocell, etc.).
[0060] The servers further include one or more universal serial bus (USB) devices (USB devices **1**-U), one or more peripheral devices (e.g., peripheral devices **1**-P), one or more memory devices (e.g., one or more flash memory devices **92**, one or more hard drive (HD) memories **94**, and one or more solid state (SS) memory devices **96**, and/or cloud memory **98**). The servers further include one or more wireless location modems **84** (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.), one or more wireless communication modems **86-1** through **86**-N(e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.), a telco interface **102** (e.g., to interface to a public switched telephone network), and a wired local area network (LAN) **88** (e.g., optical, electrical), and a wired wide area network (WAN) **90** (e.g., optical, electrical).
[0061] The computing core **52** includes a video graphics module **54**, one or more processing modules **50-1** through **50**-N(e.g., which may include one or more secure co-processors), a memory controller **56** and one or more main memories **58-1** through **58**-N(e.g., RAM serving as local memory). The computing core **52** further includes one or more input/output (I/O) device interfaces **62**, an input/output (I/O) controller **60**, a peripheral interface **64**, one or more USB interfaces **66**, one or more network interfaces **72**, one or more memory interfaces **70**, and/or one or more peripheral device interfaces **68**.
[0062] The processing modules may be a single processing device or a plurality of processing devices where the processing device may further be referred to as one or more of a “processing circuit”, a “processor”, and/or a “processing unit”. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions.
[0063] The processing module, module, processing circuit, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, and/or processing unit. Such a memory device may be a read-only

memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network).

[0064] Further note that if the processing module, module, processing circuit, and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0065] Each of the interfaces **62**, **66**, **68**, **70**, and **72** includes a combination of hardware (e.g., connectors, wiring, etc.) and may further include operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules **50-1** through **50-**N and/or a processing circuit within the interface. Each of the interfaces couples to one or more components of the servers. For example, one of the IO device interfaces **62** couples to an audio output device **78**. As another example, one of the memory interfaces **70** couples to flash memory **92** and another one of the memory interfaces **70** couples to cloud memory **98** (e.g., an on-line storage system and/or on-line backup system). In other embodiments, the servers may include more or less devices and modules than shown in this example embodiment of the servers.

[0066] FIG. **3** is a schematic block diagram of an embodiment of the various devices of the computing system **10** of FIG. **1**, including the user devices **12-1** through **12-**N and the wireless user devices **14-1** through **14-**N. The various devices include the visual output device **74** of FIG. **2**, the user input device **76** of FIG. **2**, the audio output device **78** of FIG. **2**, the visual input device **80** of FIG. **2**, and one or more sensors **82**.

[0067] The sensor may be implemented internally and/or externally to the device. Example sensors includes a still camera, a video camera, servo motors associated with a camera, a position detector, a smoke detector, a gas detector, a motion sensor, an accelerometer, velocity detector, a compass, a gyro, a temperature sensor, a pressure sensor, an altitude sensor, a humidity detector, a moisture detector, an imaging sensor, and a biometric sensor. Further examples of the sensor include an infrared sensor, an audio sensor, an ultrasonic sensor, a proximity detector, a magnetic field detector, a biomaterial detector, a radiation detector, a weight detector, a density detector, a chemical analysis detector, a fluid flow volume sensor, a DNA reader, a wind speed sensor, a wind direction sensor, and an object detection sensor.

[0068] Further examples of the sensor include an object identifier sensor, a motion recognition detector, a battery level detector, a room temperature sensor, a sound detector, a smoke detector, an intrusion detector, a motion detector, a door position sensor, a window position sensor, and a sunlight detector. Still further sensor examples include medical category sensors including: a pulse rate monitor, a heart rhythm monitor, a breathing detector, a blood pressure monitor, a blood glucose level detector, blood type, an electrocardiogram sensor, a body mass detector, an imaging sensor, a microphone, body temperature, etc.

[0069] The various devices further include the computing core **52** of FIG. **2**, the one or more universal serial bus (USB) devices (USB devices **1**-U) of FIG. **2**, the one or more peripheral devices (e.g., peripheral devices **1**-P) of FIG. **2**, and the one or more memories of FIG. **2** (e.g., flash memories **92**, HD memories **94**, SS memories **96**, and/or cloud memories **98**). The various devices further include the one or more wireless location modems **84** of FIG. **2**, the one or more wireless communication modems **86-1** through **86**-N of FIG. **2**, the telco interface **102** of FIG. **2**, the wired local area network (LAN) **88** of FIG. **2**, and the wired wide area network (WAN) **90** of FIG. **2**. In other embodiments, the various devices may include more or less internal devices and modules than shown in this example embodiment of the various devices.

[0070] FIGS. **4**A and **4**B are schematic block diagrams of another embodiment of a computing system that includes one or more of the user device **12-1** of FIG. **1**, the wireless user device **14-1** of FIG. **1**, the content source **16-1** of FIG. **1**, the transactional server **18-1** of FIG. **1**, the user device **12-2** of FIG. **1**, and the AI server **20-1** of FIG. **1**. The AI server **20-1** includes the processing module **50-1** (e.g., associated with the servers) of FIG. **2**, where the processing module **50-1** includes a collections module **120**, an identigen entigen intelligence (IEI) module **122**, and a query module **124**. Alternatively, the collections module **120**, the IEI module **122**, and the query module **124** may be implemented by the processing module **50-1** (e.g., associated with the various user devices) of FIG. **3**. The computing system functions to interpret content to produce a response to a query.

[0071] FIG. **4**A illustrates an example of the interpreting of the content to produce the response to the query where the collections module **120** interprets (e.g., based on an interpretation approach such as rules) at least one of a collections request **132** from the query module **124** and a collections request within collections information **130** from the IEI module **122** to produce content request information (e.g., potential sources, content descriptors of desired content). Alternatively, or in addition to, the collections module **120** may facilitate gathering further content based on a plurality of collection requests from a plurality of devices of the computing system **10** of FIG. **1**.

[0072] The collections request **132** is utilized to facilitate collection of content, where the content may be received in a real-time fashion once or at desired intervals, or in a static fashion from previous discrete time frames. For instance, the query module **124** issues the collections request **132** to facilitate collection of content as a background activity to support a long-term query (e.g., how many domestic airline flights over the next seven days include travelers between the age of 18 and 35 years old). The collections request **132** may include one or more of a requester identifier (ID), a content type (e.g., language, dialect, media type, topic, etc.), a content source indicator, security credentials (e.g., an authorization level, a password, a user ID, parameters utilized for encryption, etc.), a desired content quality level, trigger information (e.g., parameters under which to collect content based on a pre-event, an event (i.e., content quality level reaches a threshold to cause the trigger, trueness), or a timeframe), a desired format, and a desired timing associated with the content.

[0073] Having interpreted the collections request **132**, the collections module **120** selects a source of content based on the content request information. The selecting includes one or more of identifying one or more potential sources based on the content request information, selecting the source of content from the potential sources utilizing a selection approach (e.g., favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). For example, the collections module **120** selects the content source **16-1** when the content source **16-1** is known to provide a favorable content quality level for a domain associated with the collections request **132**.

[0074] Having selected the source of content, the collections module **120** issues a content request **126** to the selected source of content. The issuing includes generating the content request **126** based on the content request information for the selected source of content and sending the content request **126** to the selected source of content. The content request **126** may include one or more of a content type indicator, a requester ID, security credentials for content access, and any other information associated with the collections request **132**. For example, the collections module **120** sends the content request **126**, via the core network **24** of FIG. **1**, to the content source **16-1**. Alternatively, or in addition to, the collections module **120** may send a similar content request **126** to one or more of the user device **12-1**, the wireless user device **14-1**, and the transactional server

18-1 to facilitate collecting of further content.

[0075] In response to the content request **126**, the collections module **120** receives one or more content responses **128**. The content response **128** includes one or more of content associated with the content source, a content source identifier, security credential processing information, and any other information pertaining to the desired content. Having received the content response **128**, the collections module **120** interprets the received content response **128** to produce collections information **130**, where the collections information **130** further includes a collections response from the collections module **120** to the IEI module **122**.

[0076] The collections response includes one or more of transformed content (e.g., completed sentences and paragraphs), timing information associated with the content, a content source ID, and a content quality level. Having generated the collections response of the collections information **130**, the collections module **120** sends the collections information **130** to the IEI module **122**. Having received the collections information **130** from the collections module **120**, the IEI module **122** interprets the further content of the content response to generate further knowledge, where the further knowledge is stored in a memory associated with the IEI module **122** to facilitate subsequent answering of questions posed in received queries.

[0077] FIG. **4**B further illustrates the example of the interpreting of the content to produce the response to the query where, the query module **124** interprets a received query request **136** from a requester to produce an interpretation of the query request. For example, the query module **124** receives the query request **136** from the user device **12-2**, and/or from one or more of the wireless user device **14-2** and the transactional server **18-2**. The query request **136** includes one or more of an identifier (ID) associated with the request (e.g., requester ID, ID of an entity to send a response to), a question, question constraints (e.g., within a timeframe, within a geographic area, within a domain of knowledge, etc.), and content associated with the question (e.g., which may be analyzed for new knowledge itself).

[0078] The interpreting of the query request **136** includes determining whether to issue a request to the IEI module **122** (e.g., a question, perhaps with content) and/or to issue a request to the collections module **120** (e.g., for further background content). For example, the query module **124** produces the interpretation of the query request to indicate to send the request directly to the IEI module **122** when the question is associated with a simple non-time varying function answer (e.g., question: “how many hydrogen atoms does a molecule of water have?”).

[0079] Having interpreted the query request **136**, the query module **124** issues at least one of an IEI request as query information **138** to the IEI module **122** (e.g., when receiving a simple new query request) and a collections request **132** to the collections module **120** (e.g., based on two or more query requests **136** requiring more substantive content gathering). The IEI request of the query information **138** includes one or more of an identifier (ID) of the query module **124**, an ID of the requester (e.g., the user device **12-2**), a question (e.g., with regards to content for analysis, with regards to knowledge minded by the AI server from general content), one or more constraints (e.g., assumptions, restrictions, etc.) associated with the question, content for analysis of the question, and timing information (e.g., a date range for relevance of the question).

[0080] Having received the query information **138** that includes the IEI request from the query module **124**, the IEI module **122** determines whether a satisfactory response can be generated based on currently available knowledge, including that of the query request **136**. The determining includes indicating that the satisfactory response cannot be generated when an estimated quality level of an answer falls below a minimum quality threshold level. When the satisfactory response cannot be generated, the IEI module **122** facilitates collecting more content. The facilitating includes issuing a collections request to the collections module **120** of the AI server **20-1** and/or to another server or user device, and interpreting a subsequent collections response **134** of collections information **130** that includes further content to produce further knowledge to enable a more favorable answer.

[0081] When the IEI module **122** indicates that the satisfactory response can be generated, the IEI module **122** issues an IEI response as query information **138** to the query module **124**. The IEI response includes one or more of one or more answers, timing relevance of the one or more answers, an estimated quality level of each answer, and one or more assumptions associated with the answer. The issuing includes generating the IEI response based on the collections response **134** of the collections information **130** and the IEI request, and sending the IEI response as the query information **138** to the query module **124**. Alternatively, or in addition to, at least some of the further content collected by the collections module **120** is utilized to generate a collections response **134** issued by the collections module **120** to the query module **124**. The collections response **134** includes one or more of further content, a content availability indicator (e.g., when, where, required credentials, etc.), a content freshness indicator (e.g., timestamps, predicted time availability), content source identifiers, and a content quality level.
[0082] Having received the query information **138** from the IEI module **122**, the query module **124** issues a query response **140** to the requester based on the IEI response and/or the collections response **134** directly from the collections module **120**, where the collection module **120** generates the collections response **134** based on collected content and the collections request **132**. The query response **140** includes one or more of an answer, answer timing, an answer quality level, and answer assumptions.
[0083] FIG. **4**C is a logic diagram of an embodiment of a method for interpreting content to produce a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-3**, **4**A-**4**B, and also FIG. **4**C. The method includes step **150** where a collections module of a processing module of one or more computing devices (e.g., of one or more servers) interprets a collections request to produce content request information. The interpreting may include one or more of identifying a desired content source, identifying a content type, identifying a content domain, and identifying content timing requirements.
[0084] The method continues at step **152** where the collections module selects a source of content based on the content request information. For example, the collections module identifies one or more potential sources based on the content request information and selects the source of content from the potential sources utilizing a selection approach (e.g., based on one or more of favorable history, a favorable security level, favorable accessibility, favorable cost, favorable performance, etc.). The method continues at step **154** where the collections module issues a content request to the selected source of content. The issuing includes generating a content request based on the content request information for the selected source of content and sending the content request to the selected source of content.
[0085] The method continues at step **156** where the collections module issues collections information to an identigen entigen intelligence (IEI) module based on a received content response, where the IEI module extracts further knowledge from newly obtained content from the one or more received content responses. For example, the collections module generates the collections information based on newly obtained content from the one or more received content responses of the selected source of content.
[0086] The method continues at step **158** where a query module interprets a received query request from a requester to produce an interpretation of the query request. The interpreting may include determining whether to issue a request to the IEI module (e.g., a question) or to issue a request to the collections module to gather further background content. The method continues at step **160** where the query module issues a further collections request. For example, when receiving a new query request, the query module generates a request for the IEI module. As another example, when receiving a plurality of query requests for similar questions, the query module generates a request for the collections module to gather further background content.
[0087] The method continues at step **162** where the IEI module determines whether a satisfactory

query response can be generated when receiving the request from the query module. For example, the IEI module indicates that the satisfactory query response cannot be generated when an estimated quality level of an answer is below a minimum answer quality threshold level. The method branches to step **166** when the IEI module determines that the satisfactory query response can be generated. The method continues to step **164** when the IEI module determines that the satisfactory query response cannot be generated. When the satisfactory query response cannot be generated, the method continues at step **164** where the IEI module facilitates collecting more content. The method loops back to step **150**.

[0088] When the satisfactory query response can be generated, the method continues at step **166** where the IEI module issues an IEI response to the query module. The issuing includes generating the IEI response based on the collections response and the IEI request, and sending the IEI response to the query module. The method continues at step **168** where the query module issues a query response to the requester. For example, the query module generates the query response based on the IEI response and/or a collections response from the collections module and sends the query response to the requester, where the collections module generates the collections response based on collected content and the collections request.

[0089] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0090] FIG. **5**A is a schematic block diagram of an embodiment of the collections module **120** of FIG. **4**A that includes a content acquisition module **180**, a content selection module **182**, a source selection module **184**, a content security module **186**, an acquisition timing module **188**, a content transformation module **190**, and a content quality module **192**. Generally, an embodiment of this invention presents solutions where the collections module **120** supports collecting content.

[0091] In an example of operation of the collecting of the content, the content acquisition module **180** receives a collections request **132** from a requester. The content acquisition module **180** obtains content selection information **194** based on the collections request **132**. The content selection information **194** includes one or more of content requirements, a desired content type indicator, a desired content source identifier, a content type indicator, a candidate source identifier (ID), and a content profile (e.g., a template of typical parameters of the content). For example, the content acquisition module **180** receives the content selection information **194** from the content selection module **182**, where the content selection module **182** generates the content selection information **194** based on a content selection information request from the content acquisition module **180** and where the content acquisition module **180** generates the content selection information request based on the collections request **132**.

[0092] The content acquisition module **180** obtains source selection information **196** based on the collections request **132**. The source selection information **196** includes one or more of candidate source identifiers, a content profile, selected sources, source priority levels, and recommended source access timing. For example, the content acquisition module **180** receives the source selection information **196** from the source selection module **184**, where the source selection module **184** generates the source selection information **196** based on a source selection information request from the content acquisition module **180** and where the content acquisition module **180** generates the source selection information request based on the collections request **132**.

[0093] The content acquisition module **180** obtains acquisition timing information **200** based on the collections request **132**. The acquisition timing information **200** includes one or more of recommended source access timing, confirmed source access timing, source access testing results, estimated velocity of content update's, content precious, timestamps, predicted time availability, required content acquisition triggers, content acquisition trigger detection indicators, and a duplicative indicator with a pending content request. For example, the content acquisition module **180** receives the acquisition timing information **200** from the acquisition timing module **188**, where the acquisition timing module **188** generates the acquisition timing information **200** based on an acquisition timing information request from the content acquisition module **180** and where the content acquisition module **180** generates the acquisition timing information request based on the collections request **132**.

[0094] Having obtained the content selection information **194**, the source selection information **196**, and the acquisition timing information **200**, the content acquisition module **180** issues a content request **126** to a content source utilizing security information **198** from the content security module **186**, where the content acquisition module **180** generates the content request **126** in accordance with the content selection information **194**, the source selection information **196**, and the acquisition timing information **200**. The security information **198** includes one or more of source priority requirements, requester security information, available security procedures, and security credentials for trust and/or encryption. For example, the content acquisition module **180** generates the content request **126** to request a particular content type in accordance with the content selection information **194** and to include security parameters of the security information **198**, initiates sending of the content request **126** in accordance with the acquisition timing information **200**, and sends the content request **126** to a particular targeted content source in accordance with the source selection information **196**.

[0095] In response to receiving a content response **128**, the content acquisition module **180** determines the quality level of received content extracted from the content response **128**. For example, the content acquisition module **180** receives content quality information **204** from the content quality module **192**, where the content quality module **192** generates the quality level of the received content based on receiving a content quality request from the content acquisition module **180** and where the content acquisition module **180** generates the content quality request based on content extracted from the content response **128**. The content quality information includes one or more of a content reliability threshold range, a content accuracy threshold range, a desired content quality level, a predicted content quality level, and a predicted level of trust.

[0096] When the quality level is below a minimum desired quality threshold level, the content acquisition module **180** facilitates acquisition of further content. The facilitating includes issuing another content request **126** to a same content source and/or to another content source to receive and interpret further received content. When the quality level is above the minimum desired quality threshold level, the content acquisition module **180** issues a collections response **134** to the requester. The issuing includes processing the content in accordance with a transformation approach to produce transformed content, generating the collections response **134** to include the transformed content, and sending the collections response **134** to the requester. The processing of the content to produce the transformed content includes receiving content transformation information **202** from the content transformation module **190**, where the content transformation module **190** transforms the content in accordance with the transformation approach to produce the transformed content. The content transformation information includes a desired format, available formats, recommended formatting, the received content, transformation instructions, and the transformed content.

[0097] FIG. **5**B is a logic diagram of an embodiment of a method for obtaining content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-3**, **4**A-**4**C, **5**A, and also FIG. **5**B. The method includes step **210** where a processing module of one or more processing modules of one or more computing devices of the computing system receives a collections request from the requester. The method continues at step **212** where the processing module determines content selection information. The determining includes interpreting the collections request to identify requirements of the content.

[0098] The method continues at step **214** where the processing module determines source selection information. The determining includes interpreting the collections request to identify and select one or more sources for the content to be collected. The method continues at step **216** where the processing module determines acquisition timing information. The determining includes interpreting the collections request to identify timing requirements for the acquisition of the content from the one or more sources. The method continues at step **218** where the processing module issues a content request utilizing security information and in accordance with one or more of the content selection information, the source selection information, and the acquisition timing information. For example, the processing module issues the content request to the one or more sources for the content in accordance with the content requirements, where the sending of the request is in accordance with the acquisition timing information.

[0099] The method continues at step **220** where the processing module determines a content quality level for received content area. The determining includes receiving the content from the one or more sources, obtaining content quality information for the received content based on a quality analysis of the received content. The method branches to step **224** when the content quality level is favorable and the method continues to step **222** when the quality level is unfavorable. For example, the processing module determines that the content quality level is favorable when the content quality level is equal to or above a minimum quality threshold level and determines that the content quality level is unfavorable when the content quality level is less than the minimum quality threshold level.

[0100] When the content quality level is unfavorable, the method continues at step **222** where the processing module facilitates acquisition and further content. For example, the processing module issues further content requests and receives further content for analysis. When the content quality level is favorable, the method continues at step **224** where the processing module issues a collections response to the requester. The issuing includes generating the collections response and sending the collections response to the requester. The generating of the collections response may include transforming the received content into transformed content in accordance with a transformation approach (e.g., reformatting, interpreting absolute meaning and translating into another language in accordance with the absolute meaning, etc.).

[0101] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0102] FIG. **5**C is a schematic block diagram of an embodiment of the query module **124** of FIG. **4**A that includes an answer acquisition module **230**, a content requirements module **232** a source requirements module **234**, a content security module **236**, an answer timing module **238**, an answer transformation module **240**, and an answer quality module **242**. Generally, an embodiment of this invention presents solutions where the query module **124** supports responding to a query.

[0103] In an example of operation of the responding to the query, the answer acquisition module **230** receives a query request **136** from a requester. The answer acquisition module **230** obtains content requirements information **248** based on the query request **136**. The content requirements information **248** includes one or more of content parameters, a desired content type, a desired content source if any, a content type if any, candidate source identifiers, a content profile, and a question of the query request **136**. For example, the answer acquisition module **230** receives the content requirements information **248** from the content requirements module **232**, where the content requirements module **232** generates the content requirements information **248** based on a content requirements information request from the answer acquisition module **230** and where the answer acquisition module **230** generates the content requirements information request based on the query request **136**.

[0104] The answer acquisition module **230** obtains source requirements information **250** based on the query request **136**. The source requirements information **250** includes one or more of candidate source identifiers, a content profile, a desired source parameter, recommended source parameters, source priority levels, and recommended source access timing. For example, the answer acquisition module **230** receives the source requirements information **250** from the source requirements module **234**, where the source requirements module **234** generates the source requirements information **250** based on a source requirements information request from the answer acquisition module **230** and where the answer acquisition module **230** generates the source requirements information request based on the query request **136**.

[0105] The answer acquisition module **230** obtains answer timing information **254** based on the query request **136**. The answer timing information **254** includes one or more of requested answer timing, confirmed answer timing, source access testing results, estimated velocity of content updates, content freshness, timestamps, predicted time available, requested content acquisition trigger, and a content acquisition trigger detected indicator. For example, the answer acquisition module **230** receives the answer timing information **254** from the answer timing module **238**, where the answer timing module **238** generates the answer timing information **254** based on an answer timing information request from the answer acquisition module **230** and where the answer acquisition module **230** generates the answer timing information request based on the query request **136**.

[0106] Having obtained the content requirements information **248**, the source requirements information **250**, and the answer timing information **254**, the answer acquisition module **230** determines whether to issue an IEI request **244** and/or a collections request **132** based on one or more of the content requirements information **248**, the source requirements information **250**, and the answer timing information **254**. For example, the answer acquisition module **230** selects the IEI request **244** when an immediate answer to a simple query request **136** is required and is expected to have a favorable quality level. As another example, the answer acquisition module **230** selects the collections request **132** when a longer-term answer is required as indicated by the answer timing information to before and/or when the query request **136** has an unfavorable quality level.

[0107] When issuing the IEI request **244**, the answer acquisition module **230** generates the IEI request **244** in accordance with security information **252** received from the content security module **236** and based on one or more of the content requirements information **248**, the source requirements information **250**, and the answer timing information **254**. Having generated the IEI request **244**, the answer acquisition module **230** sends the IEI request **244** to at least one IEI module.

[0108] When issuing the collections request **132**, the answer acquisition module **230** generates the collections request **132** in accordance with the security information **252** received from the content security module **236** and based on one or more of the content requirements information **248**, the source requirements information **250**, and the answer timing information **254**. Having generated the collections request **132**, the answer acquisition module **230** sends the collections request **132** to at least one collections module. Alternatively, the answer acquisition module **230** facilitate sending of the collections request **132** to one or more various user devices (e.g., to access a subject matter expert).

[0109] The answer acquisition module **230** determines a quality level of a received answer extracted from a collections response **134** and/or an IEI response **246**. For example, the answer acquisition module **230** extracts the quality level of the received answer from answer quality information **258** received from the answer quality module **242** in response to an answer quality request from the answer acquisition module **230**. When the quality level is unfavorable, the answer acquisition module **230** facilitates obtaining a further answer. The facilitation includes issuing at least one of a further IEI request **244** and a further collections request **132** to generate a further answer for further quality testing. When the quality level is favorable, the answer acquisition module **230** issues a query response **140** to the requester. The issuing includes generating the query response **140** based on answer transformation information **256** received from the answer transformation module **240**, where the answer transformation module **240** generates the answer transformation information **256** to include a transformed answer based on receiving the answer from the answer acquisition module **230**. The answer transformation information **250 6**A further include the question, a desired format of the answer, available formats, recommended formatting, received IEI responses, transformation instructions, and transformed IEI responses into an answer.
[0110] FIG. **5**D is a logic diagram of an embodiment of a method for providing a response to a query within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-3**, **4**A-**4**C, **5**C, and also FIG. **5**D. The method includes step **270** where a processing module of one or more processing modules of one or more computing devices of the computing system receives a query request (e.g., a question) from a requester. The method continues at step **272** where the processing module determines content requirements information. The determining includes interpreting the query request to produce the content requirements. The method continues at step **274** where the processing module determines source requirements information. The determining includes interpreting the query request to produce the source requirements. The method continues at step **276** where the processing module determines answer timing information. The determining includes interpreting the query request to produce the answer timing information.
[0111] The method continues at step **278** the processing module determines whether to issue an IEI request and/or a collections request. For example, the determining includes selecting the IEI request when the answer timing information indicates that a simple one-time answer is appropriate. As another example, the processing module selects the collections request when the answer timing information indicates that the answer is associated with a series of events over an event time frame.
[0112] When issuing the IEI request, the method continues at step **280** where the processing module issues the IEI request to an IEI module. The issuing includes generating the IEI request in accordance with security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information.
[0113] When issuing the collections request, the method continues at step **282** where the processing module issues the collections request to a collections module. The issuing includes generating the collections request in accordance with the security information and based on one or more of the content requirements information, the source requirements information, and the answer timing information. Alternatively, the processing module issues both the IEI request and the collections request when a satisfactory partial answer may be provided based on a corresponding IEI response and a further more generalized and specific answer may be provided based on a corresponding collections response and associated further IEI response.
[0114] The method continues at step **284** where the processing module determines a quality level of a received answer. The determining includes extracting the answer from the collections response and/or the IEI response and interpreting the answer in accordance with one or more of the content requirements information, the source requirements information, the answer timing information, and the query request to produce the quality level. The method branches to step **288** when the quality level is favorable and the method continues to step **286** when the quality level is unfavorable. For

example, the processing module indicates that the quality level is favorable when the quality level is equal to or greater than a minimum answer quality threshold level. As another example, the processing module indicates that the quality level is unfavorable when the quality level is less than the minimum answer quality threshold level.

[0115] When the quality level is unfavorable, the method continues at step **286** where the processing module obtains a further answer. The obtaining includes at least one of issuing a further IEI request and a further collections request to facilitate obtaining of a further answer for further answer quality level testing as the method loops back to step **270**. When the quality level is favorable, the method continues at step **288** where the processing module issues a query response to the requester. The issuing includes transforming the answer into a transformed answer in accordance with an answer transformation approach (e.g., formatting, further interpretations of the virtual question in light of the answer and further knowledge) and sending the transformed answer to the requester as the query response.

[0116] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0117] FIG. **5**E is a schematic block diagram of an embodiment of the identigen entigen intelligence (IEI) module **122** of FIG. **4**A that includes a content ingestion module **300**, an element identification module **302**, and interpretation module **304**, and answer resolution module **306**, and an IEI control module **308**. Generally, an embodiment of this invention presents solutions where the IEI module **122** supports interpreting content to produce knowledge that may be utilized to answer questions.

[0118] In an example of operation of the producing and utilizing of the knowledge, the content ingestion module **300** generates formatted content **314** based on question content **312** and/or source content **310**, where the IEI module **122** receives an IEI request **244** that includes the question content **312** and the IEI module **122** receives a collections response **134** that includes the source content **310**. The source content **310** includes content from a source extracted from the collections response **134**. The question content **312** includes content extracted from the IEI request **244** (e.g., content paired with a question). The content ingestion module **300** generates the formatted content **314** in accordance with a formatting approach (e.g., creating proper sentences from words of the content). The formatted content **314** includes modified content that is compatible with subsequent element identification (e.g., complete sentences, combinations of words and interpreted sounds and/or inflection cues with temporal associations of words).

[0119] The element identification module **302** processes the formatted content **314** based on element rules **318** and an element list **332** to produce identified element information **340**. Rules **316** includes the element rules **318** (e.g., match, partial match, language translation, etc.). Lists **330** includes the element list **332** (e.g., element ID, element context ID, element usage ID, words, characters, symbols etc.). The IEI control module **308** may provide the rules **316** and the lists **330** by accessing stored data **360** from a memory associated with the IEI module **122**. Generally, an embodiment of this invention presents solutions where the stored data **360** may further include one or more of a descriptive dictionary, categories, representations of element sets, element list, sequence data, pending questions, pending request, recognized elements, unrecognized elements, errors, etc.

[0120] The identified element information **340** includes one or more of identifiers of elements identified in the formatted content **314**, may include ordering and/or sequencing and grouping information. For example, the element identification module **302** compares elements of the formatted content **314** to known elements of the element list **332** to produce identifiers of the known elements as the identified element information **340** in accordance with the element rules **318**. Alternatively, the element identification module **302** outputs un-identified element information **342** to the IEI control module **308**, where the un-identified element information **342** includes temporary identifiers for elements not identifiable from the formatted content **314** when compared to the element list **332**.

[0121] The interpretation module **304** processes the identified element information **340** in accordance with interpretation rules **320** (e.g., potentially valid permutations of various combinations of identified elements), question information **346** (e.g., a question extracted from the IEI request **244** which may be paired with content associated with the question), and a groupings list **334** (e.g., representations of associated groups of representations of things, a set of element identifiers, valid element usage IDs in accordance with similar, an element context, permutations of sets of identifiers for possible interpretations of a sentence or other) to produce interpreted information **344**. The interpreted information **344** includes potentially valid interpretations of combinations of identified elements. Generally, an embodiment of this invention presents solutions where the interpretation module **304** supports producing the interpreted information **344** by considering permutations of the identified element information **340** in accordance with the interpretation rules **320** and the groupings list **334**.

[0122] The answer resolution module **306** processes the interpreted information **344** based on answer rules **322** (e.g., guidance to extract a desired answer), the question information **346**, and inferred question information **352** (e.g., posed by the IEI control module or analysis of general collections of content or refinement of a stated question from a request) to produce preliminary answers **354** and an answer quality level **356**. The answer generally lies in the interpreted information **344** as both new content received and knowledge based on groupings list **334** generated based on previously received content. The preliminary answers **354** includes an answer to a stated or inferred question that subject further refinement. The answer quality level **356** includes a determination of a quality level of the preliminary answers **354** based on the answer rules **322**. The inferred question information **352** may further be associated with time information **348**, where the time information includes one or more of current real-time, a time reference associated with entity submitting a request, and a time reference of a collections response.

[0123] When the IEI control module **308** determines that the answer quality level **356** is below an answer quality threshold level, the IEI control module **308** facilitates collecting of further content (e.g., by issuing a collections request **132** and receiving corresponding collections responses **134** for analysis). When the answer quality level **356** compares favorably to the answer quality threshold level, the IEI control module **308** issues an IEI response **246** based on the preliminary answers **354**. When receiving training information **358**, the IEI control module **308** facilitates updating of one or more of the lists **330** and the rules **316** and stores the updated list **330** and the updated rules **316** in the memories as updated stored data **360**.

[0124] FIG. **5**F is a logic diagram of an embodiment of a method for analyzing content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1**-**3**, **4**A-**4**C, **5**E, and also FIG. **5**F. The method includes step **370** where a processing module of one or more processing modules of one or more computing devices of the computing system facilitates updating of one or more rules and lists based on one or more of received training information and received content. For example, the processing module updates rules with received rules to produce updated rules and updates element lists with received elements to produce updated element lists. As another example, the processing module interprets the received content to identify a new word for at least temporary inclusion in the updated element list.

[0125] The method continues at step **372** where the processing module transforms at least some of the received content into formatted content. For example, the processing module processes the received content in accordance with a transformation approach to produce the formatted content, where the formatted content supports compatibility with subsequent element identification (e.g., typical sentence structures of groups of words).
[0126] The method continues at step **374** where the processing module processes the formatted content based on the rules and the lists to produce identified element information and/or an identified element information. For example, the processing module compares the formatted content to element lists to identify a match producing identifiers for identified elements or new identifiers for unidentified elements when there is no match.
[0127] The method continues at step **376** with a processing module processes the identified element information based on rules, the lists, and question information to produce interpreted information. For example, the processing module compares the identified element information to associated groups of representations of things to generate potentially valid interpretations of combinations of identified elements.
[0128] The method continues at step **378** where the processing module processes the interpreted information based on the rules, the question information, and inferred question information to produce preliminary answers. For example, the processing module matches the interpreted information to one or more answers (e.g., embedded knowledge based on a fact base built from previously received content) with highest correctness likelihood levels that is subject to further refinement.
[0129] The method continues at step **380** where the processing module generates an answer quality level based on the preliminary answers, the rules, and the inferred question information. For example, the processing module predicts the answer correctness likelihood level based on the rules, the inferred question information, and the question information. The method branches to step **384** when the answer quality level is favorable and the method continues to step **382** when the answer quality level is unfavorable. For example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is favorable when the answer quality level is greater than or equal to a minimum answer quality threshold level. As another example, the generating of the answer quality level further includes the processing module indicating that the answer quality level is unfavorable when the answer quality level is less than the minimum answer quality threshold level.
[0130] When the answer quality level is unfavorable, the method continues at step **382** where the processing module facilitates gathering clarifying information. For example, the processing module issues a collections request to facilitate receiving further content and or request question clarification from a question requester. When the answer quality level is favorable, the method continues at step **384** where the processing module issues a response that includes one or more answers based on the preliminary answers and/or further updated preliminary answers based on gathering further content. For example, the processing module generates a response that includes one or more answers and the answer quality level and issues the response to the requester.
[0131] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.
[0132] FIG. **6**A is a schematic block diagram of an embodiment of the element identification

module **302** of FIG. **5**A and the interpretation module **304** of FIG. **5**A. The element identification module **302** includes an element matching module **400** and an element grouping module **402**. The interpretation module **304** includes a grouping matching module **404** and a grouping interpretation module **406**. Generally, an embodiment of this invention presents solutions where the element identification module **302** supports identifying potentially valid permutations of groupings of elements while the interpretation module **304** interprets the potentially valid permutations of groupings of elements to produce interpreted information that includes the most likely of groupings based on a question.

[0133] In an example of operation of the identifying of the potentially valid permutations of groupings of elements, when matching elements of the formatted content **314**, the element matching module **400** generates matched elements **412** (e.g., identifiers of elements contained in the formatted content **314**) based on the element list **332**. For example, the element matching module **400** matches a received element to an element of the element list **332** and outputs the matched elements **412** to include an identifier of the matched element. When finding elements that are unidentified, the element matching module **400** outputs un-recognized words information **408** (e.g., words not in the element list **332**, may temporarily add) as part of un-identified element information **342**. For example, the element matching module **400** indicates that a match cannot be made between a received element of the formatted content **314**, generates the unrecognized words info **408** to include the received element and/or a temporary identifier, and issues and updated element list **414** that includes the temporary identifier and the corresponding unidentified received element.

[0134] The element grouping module **402** analyzes the matched elements **412** in accordance with element rules **318** to produce grouping error information **410** (e.g., incorrect sentence structure indicators) when a structural error is detected. The element grouping module **402** produces identified element information **340** when favorable structure is associated with the matched elements in accordance with the element rules **318**. The identified element information **340** may further include grouping information of the plurality of permutations of groups of elements (e.g., several possible interpretations), where the grouping information includes one or more groups of words forming an associated set and/or super-group set of two or more subsets when subsets share a common core element.

[0135] In an example of operation of the interpreting of the potentially valid permutations of groupings of elements to produce the interpreted information, the grouping matching module **404** analyzes the identified element information **340** in accordance with a groupings list **334** to produce validated groupings information **416**. For example, the grouping matching module **404** compares a grouping aspect of the identified element information **340** (e.g., for each permutation of groups of elements of possible interpretations), generates the validated groupings information **416** to include identification of valid permutations aligned with the groupings list **334**. Alternatively, or in addition to, the grouping matching module **404** generates an updated groupings list **418** when determining a new valid grouping (e.g., has favorable structure and interpreted meaning) that is to be added to the groupings list **334**.

[0136] The grouping interpretation module **406** interprets the validated groupings information **416** based on the question information **346** and in accordance with the interpretation rules **320** to produce interpreted information **344** (e.g., most likely interpretations, next most likely interpretations, etc.). For example, the grouping interpretation module **406** obtains context, obtains favorable historical interpretations, processes the validated groupings based on interpretation rules **320**, where each interpretation is associated with a correctness likelihood level.

[0137] FIG. **6**B is a logic diagram of an embodiment of a method for interpreting information within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-3**, **4**A-**4**C, **5**E-**5**F, **6**A, and also FIG. **6**B. The method includes step **430** where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes formatted content. For example, the processing module attempt to match a received element of the formatted content to one or more elements of an elements list. When there is no match, the method branches to step **434** and when there is a match, the method continues to step **432**. When there is a match, the method continues at step **432** where the processing module outputs matched elements (e.g., to include the matched element and/or an identifier of the matched element). When there is no match, the method continues at step **434** where the processing module outputs unrecognized words (e.g., elements and/or a temporary identifier for the unmatched element).

[0138] The method continues at step **436** where the processing module analyzes matched elements. For example, the processing module attempt to match a detected structure of the matched elements (e.g., chained elements as in a received sequence) to favorable structures in accordance with element rules. The method branches to step **440** when the analysis is unfavorable and the method continues to step **438** when the analysis is favorable. When the analysis is favorable matching a detected structure to the favorable structure of the element rules, the method continues at step **438** where the processing module outputs identified element information (e.g., an identifier of the favorable structure, identifiers of each of the detected elements). When the analysis is unfavorable matching a detected structure to the favorable structure of the element rules, the method continues at step **440** where the processing module outputs grouping error information (e.g., a representation of the incorrect structure, identifiers of the elements of the incorrect structure, a temporary new identifier of the incorrect structure).

[0139] The method continues at step **442** where the processing module analyzes the identified element information to produce validated groupings information. For example, the processing module compares a grouping aspect of the identified element information and generates the validated groupings information to include identification of valid permutations that align with the groupings list. Alternatively, or in addition to, the processing module generates an updated groupings list when determining a new valid grouping.

[0140] The method continues at step **444** where the processing module interprets the validated groupings information to produce interpreted information. For example, the processing module obtains one or more of context and historical interpretations and processes the validated groupings based on interpretation rules to generate the interpreted information, where each interpretation is associated with a correctness likelihood level (e.g., a quality level).

[0141] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0142] FIG. **6**C is a schematic block diagram of an embodiment of the answer resolution module **306** of FIG. **5**A that includes an interim answer module **460**, and answer prioritization module **462**, and a preliminary answer quality module **464**. Generally, an embodiment of this invention presents solutions where the answer resolution module **306** supports producing an answer for interpreted information **344**.

[0143] In an example of operation of the providing of the answer, the interim answer module **460** analyzes the interpreted information **344** based on question information **346** and inferred question information **352** to produce interim answers **466** (e.g., answers to stated and/or inferred questions without regard to rules that is subject to further refinement). The answer prioritization module **462** analyzes the interim answers **466** based on answer rules **322** to produce preliminary answer **354**.

For example, the answer prioritization module **462** identifies all possible answers from the interim answers **466** that conform to the answer rules **322**.

[0144] The preliminary answer quality module **464** analyzes the preliminary answers **354** in accordance with the question information **346**, the inferred question information **352**, and the answer rules **322** to produce an answer quality level **356**. For example, for each of the preliminary answers **354**, the preliminary answer quality module **464** may compare a fit of the preliminary answer **354** to a corresponding previous answer and question quality level, calculate the answer quality level **356** based on a level of conformance to the answer rules **322**, calculate the answer quality level **356** based on alignment with the inferred question information **352**, and determine the answer quality level **356** based on an interpreted correlation with the question information **346**.

[0145] FIG. **6**D is a logic diagram of an embodiment of a method for producing an answer within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1**-**3**, **4**A-**4**C, **5**E-**5**F, **6**C, and also FIG. **6**D. The method includes step **480** where a processing module of one or more processing modules of one or more computing devices of the computing system analyzes received interpreted information based on question information and inferred question information to produce one or more interim answers. For example, the processing module generates potential answers based on patterns consistent with previously produced knowledge and likelihood of correctness.

[0146] The method continues at step **482** where the processing module analyzes the one or more interim answers based on answer rules to produce preliminary answers. For example, the processing module identifies all possible answers from the interim answers that conform to the answer rules. The method continues at step **484** where the processing module analyzes the preliminary answers in accordance with the question information, the inferred question information, and the answer rules to produce an answer quality level. For example, for each of the elementary answers, the processing module may compare a fit of the preliminary answer to a corresponding previous answer-and-answer quality level, calculate the answer quality level based on performance to the answer rules, calculate answer quality level based on alignment with the inferred question information, and determine the answer quality level based on interpreted correlation with the question information.

[0147] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0148] FIG. **7**A is an information flow diagram for interpreting information within a computing system, where sets of entigens **504** are interpreted from sets of identigens **502** which are interpreted from sentences of words **500**. Such identigen entigen intelligence (IEI) processing of the words (e.g., to IEI process) includes producing one or more of interim knowledge, a preliminary answer, and an answer quality level. For example, the IEI processing includes identifying permutations of identigens of a phrase of a sentence (e.g., interpreting human expressions to produce identigen groupings for each word of ingested content), reducing the permutations of identigens (e.g., utilizing rules to eliminate unfavorable permutations), mapping the reduced permutations of identigens to at least one set of entigens (e.g., most likely identigens become the entigens) to produce the interim knowledge, processing the knowledge in accordance with a knowledge database (e.g., comparing the set of entigens to the knowledge database) to produce a preliminary answer, and generating the answer quality level based on the preliminary answer for a corresponding domain.

[0149] Human expressions are utilized to portray facts and fiction about the real world. The real-world includes items, actions, and attributes. The human expressions include textual words, textual symbols, images, and other sensorial information (e.g., sounds). It is known that many words, within a given language, can mean different things based on groupings and orderings of the words. For example, the sentences of words **500** can include many different forms of sentences that mean vastly different things even when the words are very similar.

[0150] The present invention presents solutions where the computing system **10** supports producing a computer-based representation of a truest meaning possible of the human expressions given the way that multitudes of human expressions relate to these meanings. As a first step of the flow diagram to transition from human representations of things to a most precise computer representation of the things, the computer identifies the words, phrases, sentences, etc. from the human expressions to produce the sets of identigens **502**. Each identigen includes an identifier of their meaning and an identifier of an instance for each possible language, culture, etc. For example, the words car and automobile share a common meaning identifier but have different instance identifiers since they are different words and are spelled differently. As another example, the word duck is associated both with a bird and an action to elude even though they are spelled the same. In this example the bird duck has a different meaning than the elude duck and as such each has a different meaning identifier of the corresponding identigens.

[0151] As a second step of the flow diagram to transition from human representations of things to the most precise computer representation of the things, the computer extracts meaning from groupings of the identified words, phrases, sentences, etc. to produce the sets of entigens **504**. Each entigen includes an identifier of a single conceivable and perceivable thing in space and time (e.g., independent of language and other aspects of the human expressions). For example, the words car and automobile are different instances of the same meaning and point to a common shared entigen. As another example, the word duck for the bird meaning has an associated unique entigen that is different than the entigen for the word duck for the elude meaning.

[0152] As a third step of the flow diagram to transition from human expressions of things to the most precise computer representation of the things, the computer reasons facts from the extracted meanings. For example, the computer maintains a fact-based of the valid meanings from the valid groupings or sets of entigens so as to support subsequent inferences, deductions, rationalizations of posed questions to produce answers that are aligned with a most factual view. As time goes on, and as an entigen has been identified, it can encounter an experience transformations in time, space, attributes, actions, and words which are used to identify it without creating contradictions or ever losing its identity.

[0153] FIG. **7**B is a relationship block diagram illustrating an embodiment of relationships between things **510** and representations of things **512** within a computing system. The things **510** includes conceivable and perceivable things including actions **522**, items **524**, and attributes **526**. The representation of things **512** includes representations of things used by humans **514** and representation of things used by of computing devices **516** of embodiments of the present invention. The things **510** relates to the representations of things used by humans **514** where the invention presents solutions where the computing system **10** supports mapping the representations of things used by humans **514** to the representations of things used by computing devices **516**, where the representations of things used by computing devices **516** map back to the things **510**.

[0154] The representations of things used by humans **514** includes textual words **528**, textual symbols **530**, images (e.g., non-textual) **532**, and other sensorial information **534** (e.g., sounds, sensor data, electrical fields, voice inflections, emotion representations, facial expressions, whistles, etc.). The representations of things used by computing devices **516** includes identigens **518** and entigens **520**. The representations of things used by humans **514** maps to the identigens **518** and the identigens **518** map to the entigens **520**. The entigens **520** uniquely maps back to the things **510** in space and time, a truest meaning the computer is looking for to create knowledge and answer questions based on the knowledge.

[0155] To accommodate the mapping of the representations of things used by humans **514** to the identigens **518**, the identigens **518** is partitioned into actenyms **544** (e.g., actions), itenyms **546** (e.g., items), attrenyms **548** (e.g., attributes), and functionals **550** (e.g., that join and/or describe). Each of the actenyms **544**, itenyms **546**, and attrenyms **548** may be further classified into singulatums **552** (e.g., identify one unique entigen) and pluratums **554** (e.g., identify a plurality of entigens that have similarities).

[0156] Each identigen **518** is associated with an identigens identifier (IDN) **536**. The IDN **536** includes a meaning identifier (ID) **538** portion, an instance ID **540** portion, and a type ID **542** portion. The meaning ID **538** includes an identifier of common meaning. The instance ID **540** includes an identifier of a particular word and language. The type ID **542** includes one or more identifiers for actenyms, itenyms, attrenyms, singulatums, pluratums, a time reference, and any other reference to describe the IDN **536**. The mapping of the representations of things used by humans **514** to the identigens **518** by the computing system of the present invention includes determining the identigens **518** in accordance with logic and instructions for forming groupings of words.

[0157] Generally, an embodiment of this invention presents solutions where the identigens **518** map to the entigens **520**. Multiple identigens may map to a common unique entigen. The mapping of the identigens **518** to the entigens **520** by the computing system of the present invention includes determining entigens in accordance with logic and instructions for forming groupings of identigens.

[0158] FIG. 7C is a diagram of an embodiment of a synonym words table **570** within a computing system, where the synonym words table **570** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the synonym words table **570** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the words car, automobile, auto, bil (Swedish), carro (Spanish), and bil (Danish) all share a common meaning but are different instances (e.g., different words and languages). The words map to a common meaning ID but to individual unique instant identifiers. Each of the different identigens map to a common entigen since they describe the same thing.

[0159] FIG. 7D is a diagram of an embodiment of a polysemous words table **576** within a computing system, where the polysemous words table **576** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538** and the instance ID **540**. The computing system of the present invention may utilize the polysemous words table **576** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the word duck maps to four different identigens since the word duck has four associated different meanings (e.g., bird, fabric, to submerge, to elude) and instances. Each of the identigens represent different things and hence map to four different entigens.

[0160] FIG. 7E is a diagram of an embodiment of transforming words into groupings within a computing system that includes a words table **580**, a groupings of words section to validate permutations of groupings, and a groupings table **584** to capture the valid groupings. The words table **580** includes multiple fields including textual words **572**, identigens **518**, and entigens **520**. The identigens **518** includes fields for the meaning identifier (ID) **538**, the instance ID **540**, and the type ID **542**. The computing system of the present invention may utilize the words table **580** to map textual words **572** to identigens **518** and map the identigens **518** to entigens **520**. For example, the word pilot may refer to a flyer and the action to fly. Each meaning has a different identigen and different entigen.

[0161] The computing system the present invention may apply rules to the fields of the words table **580** to validate various groupings of words. Those that are invalid are denoted with a “X” while those that are valid are associated with a check mark. For example, the grouping “pilot Tom” is invalid when the word pilot refers to flying and Tom refers to a person. The identigen combinations for the flying pilot and the person Tom are denoted as invalid by the rules. As another example, the grouping “pilot Tom” is valid when the word pilot refers to a flyer and Tom refers to the person. The identigen combinations for the flyer pilot and the person Tom are denoted as valid by the rules.

[0162] The groupings table **584** includes multiple fields including grouping ID **586**, word strings **588**, identigens **518**, and entigens **520**. The computing system of the present invention may produce the groupings table **584** as a stored fact base for valid and/or invalid groupings of words identified by their corresponding identigens. For example, the valid grouping “pilot Tom” referring to flyer Tom the person is represented with a grouping identifier of **3001** and identity and identifiers 150.001 and 457.001. The entigen field **520** may indicate associated entigens that correspond to the identigens. For example, entigen e717 corresponds to the flyer pilot meaning and entigen e61 corresponds to the time the person meaning. Alternatively, or in addition to, the entigen field **520** may be populated with a single entigen identifier (ENI).

[0163] The word strings field **588** may include any number of words in a string. Different ordering of the same words can produce multiple different strings and even different meanings and hence entigens. More broadly, each entry (e.g., role) of the groupings table **584** may refer to groupings of words, two or more word strings, an idiom, just identigens, just entigens, and/or any combination of the preceding elements. Each entry has a unique grouping identifier. An idiom may have a unique grouping ID and include identifiers of original word identigens and replacing identigens associated with the meaning of the idiom not just the meaning of the original words. Valid groupings may still have ambiguity on their own and may need more strings and/or context to select a best fit when interpreting a truest meaning of the grouping.

[0164] FIG. **8**A is a data flow diagram for accumulating knowledge within a computing system, where a computing device, at a time=t**0**, ingests and processes facts **598** at a step **590** based on rules **316** and fact base information **600** to produce groupings **602** for storage in a fact base **592** (e.g., words, phrases, word groupings, identigens, entigens, quality levels). The facts **598** may include information from books, archive data, Central intelligence agency (CIA) world fact book, trusted content, etc. The ingesting may include filtering to organize and promote better valid groupings detection (e.g., considering similar domains together). The groupings **602** includes one or more of groupings identifiers, identigen identifiers, entigen identifiers, and estimated fit quality levels. The processing step **590** may include identifying identigens from words of the facts **598** in accordance with the rules **316** and the fact base info **600** and identifying groupings utilizing identigens in accordance with rules **316** and fact base info **600**.

[0165] Subsequent to ingestion and processing of the facts **598** to establish the fact base **592**, at a time=t**1**+, the computing device ingests and processes new content **604** at a step **594** in accordance with the rules **316** and the fact base information **600** to produce preliminary grouping **606**. The new content may include updated content (e.g., timewise) from periodicals, newsfeeds, social media, etc. The preliminary grouping **606** includes one or more of preliminary groupings identifiers, preliminary identigen identifiers, preliminary entigen identifiers, estimated fit quality levels, and representations of unidentified words.

[0166] The computing device validates the preliminary groupings **606** at a step **596** based on the rules **316** and the fact base info **600** to produce updated fact base info **608** for storage in the fact base **592**. The validating includes one or more of reasoning a fit of existing fact base info **600** with the new preliminary grouping **606**, discarding preliminary groupings, updating just time frame information associated with an entry of the existing fact base info **600** (e.g., to validate knowledge for the present), creating new entigens, and creating a median entigen to summarize portions of knowledge within a median indicator as a quality level indicator (e.g., suggestive not certain).

[0167] Storage of the updated fact base information **608** captures patterns that develop by themselves instead of searching for patterns as in prior art artificial intelligence systems. Growth of the fact base **592** enables subsequent reasoning to create new knowledge including deduction, induction, inference, and inferential sentiment (e.g., a chain of sentiment sentences). Examples of sentiments includes emotion, beliefs, convictions, feelings, judgments, notions, opinions, and views.

[0168] FIG. **8**B is a diagram of an embodiment of a groupings table **620** within a computing system. The groupings table **620** includes multiple fields including grouping ID **586**, word strings **588**, an IF string **622** and a THEN string **624**. Each of the fields for the IF string **622** and the THEN string **624** includes fields for an identigen (IDN) string **626**, and an entigen (ENI) string **628**. The computing system of the present invention may produce the groupings table **620** as a stored fact base to enable IF THEN based inference to generate a new knowledge inference **630**.

[0169] As a specific example, grouping **5493** points out the logic of IF someone has a tumor, THEN someone is sick and the grouping **5494** points out the logic that IF someone is sick, THEN someone is sad. As a result of utilizing inference, the new knowledge inference **630** may produce grouping **5495** where IF someone has a tumor, THEN someone is possibly sad (e.g., or is sad).

[0170] FIG. **8**C is a data flow diagram for answering questions utilizing accumulated knowledge within a computing system, where a computing device ingests and processes question information **346** at a step **640** based on rules **316** and fact base info **600** from a fact base **592** to produce preliminary grouping **606**. The ingesting and processing questions step **640** includes identifying identigens from words of a question in accordance with the rules **316** and the fact base information **600** and may also include identifying groupings from the identified identigens in accordance with the rules **316** and the fact base information **600**.

[0171] The computing device validates the preliminary grouping **606** at a step **596** based on the rules **316** and the fact base information **600** to produce identified element information **340**. For example, the computing device reasons fit of existing fact base information with new preliminary groupings **606** to produce the identified element information **340** associated with highest quality levels. The computing device interprets a question of the identified element information **340** at a step **642** based on the rules **316** and the fact base information **600**. The interpreting of the question may include separating new content from the question and reducing the question based on the fact base information **600** and the new content.

[0172] The computing device produces preliminary answers **354** from the interpreted information **344** at a resolve answer step **644** based on the rules **316** and the fact base information **600**. For example, the computing device compares the interpreted information **344** two the fact base information **600** to produce the preliminary answers **354** with highest quality levels utilizing one or more of deduction, induction, inferencing, and applying inferential sentiments logic. Alternatively, or in addition to, the computing device may save new knowledge identified from the question information **346** to update the fact base **592**.

[0173] FIG. **8**D is a data flow diagram for answering questions utilizing interference within a computing system that includes a groupings table **648** and the resolve answer step **644** of FIG. **8**C. The groupings table **648** includes multiple fields including fields for a grouping (GRP) identifier (ID) **586**, word strings **588**, an identigen (IDN) string **626**, and an entigen (ENI) **628**. The groupings table **648** may be utilized to build a fact base to enable resolving a future question into an answer. For example, the grouping **8356** notes knowledge that Michael sleeps eight hours and grouping **8357** notes that Michael usually starts to sleep at 11 PM.

[0174] In a first question example that includes a question “Michael sleeping?”, the resolve answer step **644** analyzes the question from the interpreted information **344** in accordance with the fact base information **600**, the rules **316**, and a real-time indicator that the current time is 1 AM to produce a preliminary answer of “possibly YES” when inferring that Michael is probably sleeping at 1 AM when Michael usually starts sleeping at 11 PM and Michael usually sleeps for a duration of eight hours.

[0175] In a second question example that includes the question “Michael sleeping?”, the resolve answer step **644** analyzes the question from the interpreted information **344** in accordance with the fact base information **600**, the rules **316**, and a real-time indicator that the current time is now 11 AM to produce a preliminary answer of "possibly NO" when inferring that Michael is probably not sleeping at 11 AM when Michael usually starts sleeping at 11 PM and Michael usually sleeps for a duration of eight hours.

[0176] FIG. **8**E is a relationship block diagram illustrating another embodiment of relationships between things and representations of things within a computing system. While things in the real world are described with words, it is often the case that a particular word has multiple meanings in isolation. Interpreting the meaning of the particular word may hinge on analyzing how the word is utilized in a phrase, a sentence, multiple sentences, paragraphs, and even whole documents or more. Describing and stratifying the use of words, word types, and possible meanings help in interpreting a true meaning.

[0177] Humans utilize textual words **528** to represent things in the real world. Quite often a particular word has multiple instances of different grammatical use when part of a phrase of one or more sentences. The grammatical use **649** of words includes the nouns and the verbs, and also includes adverbs, adjectives, pronouns, conjunctions, prepositions, determiners, exclamations, etc.

[0178] As an example of multiple grammatical use, the word "bat" in the English language can be utilized as a noun or a verb. For instance, when utilized as a noun, the word "bat" may apply to a baseball bat or may apply to a flying "bat." As another instance, when utilized as a verb, the word "bat" may apply to the action of hitting or batting an object, i.e., "bat the ball."

[0179] To stratify word types by use, the words are associated with a word type (e.g., type identifier **542**). The word types include objects (e.g., items **524**), characteristics (e.g., attributes **526**), actions **522**, and the functionals **550** for joining other words and describing words. For example, when the word "bat" is utilized as a noun, the word is describing the object of either the baseball bat or the flying bat. As another example, when the word "bat" is utilized as a verb, the word is describing the action of hitting.

[0180] To determine possible meanings, the words, by word type, are mapped to associative meanings (e.g., identigens **518**). For each possible associative meaning, the word type is documented with the meaning and further with an identifier (ID) of the instance (e.g., an identigen identifier).

[0181] For the example of the word "bat" when utilized as a noun for the baseball bat, a first identigen identifier **536-1** includes a type ID **542-1** associated with the object **524**, an instance ID **540-1** associated with the first identigen identifier (e.g., unique for the baseball bat), and a meaning ID **538-1** associated with the baseball bat. For the example of the word "bat" when utilized as a noun for the flying bat, a second identigen identifier **536-2** includes a type ID **542-1** associated with the object **524**, an instance ID **540-2** associated with the second identigen identifier (e.g., unique for the flying bat), and a meaning ID **538-2** associated with the flying bat. For the example of the word "bat" when utilized as a verb for the bat that hits, a third identigen identifier **536-2** includes a type ID **542-2** associated with the actions **522**, an instance ID **540-3** associated with the third identigen identifier (e.g., unique for the bat that hits), and a meaning ID **538-3** associated with the bat that hits.

[0182] With the word described by a type and possible associative meanings, a combination of full grammatical use of the word within the phrase etc., application of rules, and utilization of an ever-growing knowledge database that represents knowledge by linked entigens, the absolute meaning (e.g., entigen **520**) of the word is represented as a unique entigen. For example, a first entigen e1 represents the absolute meaning of a baseball bat (e.g., a generic baseball bat not a particular baseball bat that belongs to anyone), a second entigen e2 represents the absolute meaning of the flying bat (e.g., a generic flying bat not a particular flying bat), and a third entigen e3 represents the absolute meaning of the verb bat (e.g., to hit).

[0183] An embodiment of methods to ingest text to produce absolute meanings for storage in a knowledge database are discussed in greater detail with reference to FIGS. **8**F-H. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to definitively interpret the absolute meaning of a string of words.

[0184] Another embodiment of methods to respond to a query to produce an answer based on knowledge stored in the knowledge database are discussed in greater detail with reference to FIGS. **8**J-L. Those embodiments further discuss the discerning of the grammatical use, the use of the rules, and the utilization of the knowledge database to interpret the query. The query interpretation is utilized to extract the answer from the knowledge database to facilitate forming the query response.

[0185] FIGS. **8**F and **8**G are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5**E, the element identification module **302** of FIG. **5**E, the interpretation module **304** of FIG. **5**E, the IEI control module **308** of FIG. **5**E, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention provides presents solutions where the computing system **10** supports processing content to produce knowledge for storage in a knowledge database.

[0186] The processing of the content to produce the knowledge includes a series of steps. For example, a first step includes identifying words of an ingested phrase to produce tokenized words. As depicted in FIG. **8**F, a specific example of the first step includes the content ingestion module **300** comparing words of source content **310** to dictionary entries to produce formatted content **314** that includes identifiers of known words. Alternatively, when a comparison is unfavorable, the temporary identifier may be assigned to an unknown word. For instance, the content ingestion module **300** produces identifiers associated with the words “the”, “black”, “bat”, “eats”, and “fruit” when the ingested phrase includes “The black bat eats fruit”, and generates the formatted content **314** to include the identifiers of the words.

[0187] A second step of the processing of the content to produce the knowledge includes, for each tokenized word, identifying one or more identigens that correspond the tokenized word, where each identigen describes one of an object, a characteristic, and an action. As depicted in FIG. **8**F, a specific example of the second step includes the element identification module **302** performing a look up of identigen identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized word of the formatted content **314** to produce identified element information **340**.

[0188] A unique identifier is associated with each of the potential object, the characteristic, and the action (OCA) associated with the tokenized word (e.g., sequential identigens). For instance, the element identification module **302** identifies a functional symbol for “the”, identifies a single identigen for “black”, identifies two identigens for “bat” (e.g., baseball bat and flying bat), identifies a single identigen for “eats”, and identifies a single identigen for “fruit.” When at least one tokenized word is associated with multiple identigens, two or more permutations of sequential combinations of identigens for each tokenized word result. For example, when “bat” is associated with two identigens, two permutations of sequential combinations of identigens result for the ingested phrase.

[0189] A third step of the processing of the content to produce the knowledge includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation), where the equation package includes a sequential linking of pairs of identigens (e.g., relationships), where each sequential linking pairs a preceding identigen to a next identigen, and where an equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc. Multiple OCAs occur for a common word when the word has multiple potential meanings (e.g., a baseball bat, a flying bat).

[0190] As depicted in FIG. **8**F, a specific example of the third step includes the interpretation module **304**, for each permutation of identigens of each tokenized word of the identified element information **340**, the interpretation module **304** generates, in accordance with interpretation rules **320** and a groupings list **334**, an equation package to include one or more of the identifiers of the tokenized words, a list of identifiers of the identigens of the equation package, a list of pairing identifiers for sequential pairs of identigens, and a quality metric associated with each sequential pair of identigens (e.g., likelihood of a proper interpretation). For instance, the interpretation module **304** produces a first equation package that includes a first identigen pairing of a black bat (e.g., flying bat with a higher quality metric level), the second pairing of bat eats (e.g., the flying bat eats, with a higher quality metric level), and a third pairing of eats fruit, and the interpretation module **304** produces a second equation package that includes a first pairing of a black bat (e.g., baseball bat, with a neutral quality metric level), the second pairing of bat eats (e.g., the baseball bat eats, with a lower quality metric level), and a third pairing of eats fruit.

[0191] A fourth step of the processing of the content to produce the knowledge includes selecting a surviving equation package associated with a most favorable confidence level. As depicted in FIG. **8**F, a specific example of the fourth step includes the interpretation module **304** applying interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce a number of permutations of the sequential combinations of identigens to produce interpreted information **344** that includes identification of at least one equation package as a surviving interpretation SI (e.g., higher quality metric level).

[0192] Non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation NSI **2** (e.g., lower quality metric level), where an overall quality metric level may be assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level of an equation package indicates a higher probability of a most favorable interpretation. For instance, the interpretation module **304** eliminates the equation package that includes the second pairing indicating that the “baseball bat eats” which is inconsistent with a desired quality metric level of one or more of the groupings list **334** and the interpretation rules **320** and selects the equation package associated with the “flying bat eats” which is favorably consistent with the one or more of the quality metric levels of the groupings list **334** and the interpretation rules **320**.

[0193] A fifth step of the processing of the content to produce the knowledge utilizing the confidence level includes integrating knowledge of the surviving equation package into a knowledge database. For example, integrating at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. As another example, the portion of the reduced OCA combinations may be translated into rows and columns entries when utilizing a rows and columns database rather than a graphical database. When utilizing the rows and columns approach for the knowledge database, subsequent access to the knowledge database may utilize structured query language (SQL) queries.

[0194] As depicted in FIG. **8**G, a specific example of the fifth step includes the IEI control module **308** recovering fact base information **600** from SS memory **96** to identify a portion of the knowledge database for potential modification utilizing the OCAs of the surviving interpretation SI **1** (i.e., compare a pattern of relationships between the OCAs of the surviving interpretation SI **1** from the interpreted information **344** to relationships of OCAs of the portion of the knowledge database including potentially new quality metric levels).

[0195] The fifth step further includes determining modifications (e.g., additions, subtractions, further clarifications required when information is complex, etc.) to the portion of the knowledge database based on the new quality metric levels. For instance, the IEI control module **308** causes adding the element “black” as a “describes” relationship of an existing bat OCA and adding the element “fruit” as an eats “does to” relationship to implement the modifications to the portion of the fact base information **600** to produce updated fact base information **608** for storage in the SS memory **96**.

[0196] FIG. **8**H is a logic diagram of an embodiment of a method for processing content to produce knowledge for storage within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1**-**8**E, **8**F, and also FIG. **8**G. The method includes step **650** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested phrase to produce tokenized words. The identified includes comparing words to known words of dictionary entries to produce identifiers of known words.

[0197] For each tokenized word, the method continues at step **651** where the processing module identifies one or more identigens that corresponds to the tokenized word, where each identigen describes one of an object, a characteristic, and an action (e.g., OCA). The identifying includes performing a lookup of identifiers of the one or more identigens associated with each tokenized word, where the different identifiers associated with each of the potential object, the characteristic, and the action associated with the tokenized word.

[0198] The method continues at step **652** where the processing module, for each permutation of sequential combinations of identigens, generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module generates the equation package to include a plurality of equation elements, where each equation element describes the relationship (e.g., describes, acts on, is a, belongs to, did, did too, etc.) between sequentially adjacent identigens of a plurality of sequential combinations of identigens. Each equation element may be further associated with a quality metric to evaluate a favorability level of an interpretation in light of the sequence of identigens of the equation package.

[0199] The method continues at step **653** where the processing module selects a surviving equation package associated with most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens), to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package, where non-surviving equation packages are eliminated that compare unfavorably to pairing rules and/or are associated with an unfavorable quality metric levels to produce a non-surviving interpretation, where an overall quality metric level is assigned to each equation package based on quality metric levels of each pairing, such that a higher quality metric level indicates an equation package with a higher probability of favorability of correctness.

[0200] The method continues at step **654** where the processing module integrates knowledge of the surviving equation package into a knowledge database. For example, the processing module integrates at least a portion of the reduced OCA combinations into a graphical database to produce updated knowledge. The integrating may include recovering fact base information from storage of the knowledge database to identify a portion of the knowledge database for potential modifications utilizing the OCAs of the surviving equation package (i.e., compare a pattern of relationships between the OCAs of the surviving equation package to relationships of the OCAs of the portion of the knowledge database including potentially new quality metric levels). The integrating further includes determining modifications (e.g., additions, subtractions, further clarifications required when complex information is presented, etc.) to produce the updated knowledge database that is based on fit of acceptable quality metric levels, and implementing the modifications to the portion of the fact base information to produce the updated fact base information for storage in the portion of the knowledge database.

[0201] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0202] FIGS. **8**J and **8**K are schematic block diagrams of another embodiment of a computing system that includes the content ingestion module **300** of FIG. **5**E, the element identification module **302** of FIG. **5**E, the interpretation module **304** of FIG. **5**E, the answer resolution module **306** of FIG. **5**E, and the SS memory **96** of FIG. **2**. Generally, an embodiment of this invention provides solutions where the computing system **10** supports for generating a query response to a query utilizing a knowledge database.

[0203] The generating of the query response to the query includes a series of steps. For example, a first step includes identifying words of an ingested query to produce tokenized words. As depicted in FIG. **8**J, a specific example of the first step includes the content ingestion module **300** comparing words of query info **138** to dictionary entries to produce formatted content **314** that includes identifiers of known words. For instance, the content ingestion module **300** produces identifiers for each word of the query "what black animal flies and eats fruit and insects?"

[0204] A second step of the generating of the query response to the query includes, for each tokenized word, identifying one or more identigens that correspond the tokenized word, where each identigen describes one of an object, a characteristic, and an action (OCA). As depicted in FIG. **8**J, a specific example of the second step includes the element identification module **302** performing a look up of identifiers, utilizing an element list **332** and in accordance with element rules **318**, of the one or more identigens associated with each tokenized word of the formatted content **314** to produce identified element information **340**. A unique identifier is associated with each of the potential object, the characteristic, and the action associated with a particular tokenized word. For instance, the element identification module **302** produces a single identigen identifier for each of the black color, an animal, flies, eats, fruit, and insects.

[0205] A third step of the generating of the query response to the query includes, for each permutation of sequential combinations of identigens, generating a corresponding equation package (i.e., candidate interpretation). The equation package includes a sequential linking of pairs of identigens, where each sequential linking pairs a preceding identigen to a next identigen. An equation element describes a relationship between paired identigens (OCAs) such as describes, acts on, is a, belongs to, did, did to, etc.

[0206] As depicted in FIG. **8**J, a specific example of the third step includes the interpretation module **304**, for each permutation of identigens of each tokenized word of the identified element information **340**, generating the equation packages in accordance with interpretation rules **320** and a groupings list **334** to produce a series of equation elements that include pairings of identigens. For instance, the interpretation module **304** generates a first pairing to describe a black animal, a second pairing to describe an animal that flies, a third pairing to describe flies and eats, a fourth pairing to describe eats fruit, and a fifth pairing to describe eats fruit and insects.

[0207] A fourth step of the generating the query response to the query includes selecting a surviving equation package associated with a most favorable interpretation. As depicted in FIG. **8**J, a specific example of the fourth step includes the interpretation module **304** applying the interpretation rules **320** (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to produce interpreted information **344**. The interpreted information **344** includes identification of at least one equation package as a surviving interpretation SI **10**, where non-surviving equation packages, if any, are eliminated that compare unfavorably to pairing rules to produce a non-surviving interpretation.

[0208] A fifth step of the generating the query response to the query includes utilizing a knowledge database, generating a query response to the surviving equation package of the query, where the surviving equation package of the query is transformed to produce query knowledge for comparison to a portion of the knowledge database. An answer is extracted from the portion of the knowledge database to produce the query response.

[0209] As depicted in FIG. **8**K, a specific example of the fifth step includes the answer resolution module **306** interpreting the surviving interpretation SI **10** of the interpreted information **344** in accordance with answer rules **322** to produce query knowledge QK **10** (i.e., a graphical representation of knowledge when the knowledge database utilizes a graphical database). For example, the answer resolution module **306** accesses fact base information **600** from the SS memory **96** to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge QK **10** (e.g., by comparing attributes of the query knowledge QK **10** to attributes of the fact base information **600**), and generates preliminary answers **354** that includes the answer to the query. For instance, the answer is "bat" when the associated OCAs of bat, such as black, eats fruit, eats insects, is an animal, and flies, aligns with OCAs of the query knowledge.

[0210] FIG. **8**L is a logic diagram of an embodiment of a method for generating a query response to a query utilizing knowledge within a knowledge database within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8**D, **8**J, and also FIG. **8**K. The method includes step **655** where a processing module of one or more processing modules of one or more computing devices of the computing system identifies words of an ingested query to produce tokenized words. For example, the processing module compares words to known words of dictionary entries to produce identifiers of known words.

[0211] For each tokenized word, the method continues at step **656** where the processing module identifies one or more identigens that correspond to the tokenized word, where each identigen describes one of an object, a characteristic, and an action. For example, the processing module performs a lookup of identifiers of the one or more identigens associated with each tokenized word, where different identifiers associated with each permutation of a potential object, characteristic, and action associated with the tokenized word.

[0212] For each permutation of sequential combinations of identigens, the method continues at step **657** where the processing module generates a plurality of equation elements to form a corresponding equation package, where each equation element describes a relationship between sequentially linked pairs of identigens. Each sequential linking pairs a preceding identigen to a next identigen. For example, for each permutation of identigens of each tokenized word, the processing module includes all other permutations of all other tokenized words to generate the equation packages. Each equation package includes a plurality of equation elements describing the relationships between sequentially adjacent identigens of a plurality of sequential combinations of identigens.

[0213] The method continues at step **658** where the processing module selects a surviving equation package associated with a most favorable interpretation. For example, the processing module applies rules (i.e., inference, pragmatic engine, utilizing the identifiers of the identigens to match against known valid combinations of identifiers of entigens) to reduce the number of permutations of the sequential combinations of identigens to identify at least one equation package. Non-surviving equation packages are eliminated that compare unfavorably to pairing rules. The method continues at step **659** where the processing module generates a query response to the surviving equation package, where the surviving equation package is transformed to produce query knowledge for locating the portion of a knowledge database that includes an answer to the query. As an example of generating the query response, the processing module interprets the surviving the equation package in accordance with answer rules to produce the query knowledge (e.g., a graphical representation of knowledge when the knowledge database utilizes a graphical database format).

[0214] The processing module accesses fact base information from the knowledge database to identify the portion of the knowledge database associated with a favorable comparison of the query knowledge (e.g., favorable comparison of attributes of the query knowledge to the portion of the knowledge database, aligning favorably comparing entigens without conflicting entigens). The processing module extracts an answer from the portion of the knowledge database to produce the query response.

[0215] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0216] FIGS. **9**A and **9**B are schematic block diagrams of another embodiment of a computing system illustrating a method for converting content of a first aptitude level to converted content of a second aptitude level within the computing system in accordance with the present invention. The computing system includes the content ingestion module **300** of FIG. **5**E, the element identification module **302** of FIG. **5**E, the interpretation module **304** of FIG. **5**E, the answer resolution module **306** of FIG. **5**E, and a knowledge database **700**. The knowledge database **700** may be implemented utilizing one or more of the memories of FIG. **2**.

[0217] FIG. **9**A illustrates an example of operation of steps of a method for converting the content of the first aptitude level to the converted content of the second aptitude level where the content ingestion module **300** partitions a first aptitude level phrase (e.g., utilizing a dictionary) to produce phrase words. For example, the content ingestion module **300** receives phrase **702** that includes the first aptitude level phrase “black bats eat fruit” and partitions the phrase **702** to produce phrase words **704** that includes the words “black”, “bats”, “eat”, and “fruit”.

[0218] The element identification module **302** identifies a set of identigens **708** for each word of the first aptitude level phrase to produce a plurality of sets of identigens. For example, the element identification module **302** accesses, for each word of the phrase words **704**, the knowledge database **700** to retrieve an identigen set. For instance, the element identification module **302** receives identigen information **706** from the knowledge database **700** that includes an identigen set #**1** for the word “black”, where the identigen set #**1** includes identigens **1**, **2**, and **3**. The identigens **1**, **2**, and **3** represent three unique interpretations of the word black, including “dark-skin people”, “black color”, and “to make black.”

[0219] The interpretation module **304** generates a first aptitude level entigen group **712** for the first aptitude level phrase in accordance with identigen rules **710**. The first aptitude level entigen group **712** represents a most likely interpretation of the first aptitude level phrase. The first aptitude level phrase and the first aptitude level entigen group are associated with a first aptitude level of a plurality of aptitude levels.

[0220] The generating of the first aptitude level entigen group **712** includes interpreting, utilizing the identigen rules **710**, the plurality of sets of identigens **708** to produce the first aptitude level entigen group **712**. A set of identigens of the plurality of sets of identigens includes one or more different meanings of a word of the first aptitude level phrase. A first aptitude level entigen of the first aptitude level entigen group corresponds to an identigen of the set of identigens having a selected meaning of the one or more different meanings of the word of the first aptitude level phrase. For example, the interpretation module **304** identifies allowed pairings of identigens to include identigens **2** and **5**, **5** and **8**, and **8** and **9** to produce the first aptitude level entigen group **712** to include entigens **2**, **5**, **8**, and **9**. Alternatively, or in addition to, the interpretation module **304** identifies disallowed pairings of identigens (e.g., **1** and **4**, **1** and **5**, etc.) to produce the first aptitude level entigen group **712** that includes entigens **2**, **5**, **8**, and **9**.

[0221] FIG. **9**B further illustrates the example of operation of steps of the method for converting the content of the first aptitude level to the converted content of the second aptitude level where the answer resolution module **306** obtains a multiple aptitude level entigen group **714** from the knowledge database **700** based on the first aptitude level entigen group **712**. The multiple aptitude level entigen group **714** substantially includes the first aptitude level entigen group **712**. The multiple aptitude level entigen group **714** is associated with the plurality of aptitude levels.

[0222] In an embodiment, higher levels of aptitude are associated with more embellishment and more knowledge around connected meanings of the multiple aptitude level entigen group **714**. The multiple aptitude level entigen group **714** includes representations of entigens and connectors that represent relationships between the entigens. The connectors describe relationships between the objects, actions, and characteristic entigens. The relationships include one or more of describes, acts on, is, is a, belongs to, did, did too, etc.

[0223] In an embodiment of the multiple aptitude level entigen group **714**, a first aptitude level includes knowledge that bats are black and eat fruit. In a second aptitude level, further knowledge represents bats can also be Brown, are mammals, and also eat insects. In a third aptitude level, even further knowledge represents bats can be flying mammals.

[0224] The obtaining of the multiple aptitude level entigen group **714** from the knowledge database **700** based on the first aptitude level entigen group **712** includes identifying a group of entigens of the knowledge database **700** that compares favorably to the first aptitude level entigen group **712** as the multiple aptitude level entigen group **714**. A first entigen of the multiple aptitude level entigen group **714** is substantially the same as a first entigen of the first aptitude level entigen group **712**. A second entigen of the multiple aptitude level entigen group **714** is substantially the same as a second entigen of the first aptitude level entigen group **712**. A first entigen relationship between the first and second entigens of the first aptitude level entigen group **712** is substantially the same as a second entigen relationship between the first and second entigens of the multiple aptitude level entigen group **714**. For instance, the answer resolution module **306** receives entigen information **716** that includes entigens **2**, **5**, **8**, and **9** of the multiple aptitude level entigen group **714** that matches the entigens of the first aptitude level entigen group **712**.

[0225] Alternatively, or in addition to, the answer resolution module **306** identifies the first aptitude level based on an aptitude level affiliation of a set of entigens of the multiple aptitude level entigen group that corresponds to the first aptitude level entigen group. For example, the answer resolution module **306** interprets the entigen information **716** to identify the entigens **2**, **5**, **8**, and **9** of the multiple aptitude level entigen group **714** as associated with the first aptitude level.

[0226] Having obtained the multiple aptitude level entigen group **714**, the answer resolution module **306** generates a second aptitude level entigen group **718** utilizing the multiple aptitude level entigen group **714**. The second aptitude level entigen group **718** is associated with a second aptitude level of the plurality of aptitude levels. The second aptitude level is different than the first aptitude level. The generating of the second aptitude level entigen group **718** includes a series of steps.

[0227] A first step includes determining the second aptitude level. The determining may be based on one or more of a higher aptitude level for more detail, a lower aptitude level for less detail, a particular entigen to be included or excluded, an entigen characteristic (i.e., color, entigen type etc.), a number of characteristic combinations (i.e., include two colors, include two types of food, etc.). For example, the answer resolution module **306** determines the second aptitude level to include one or more level of detail.

[0228] A second step of the generating of the second aptitude level entigen group **718** includes selecting entigens of the multiple aptitude level entigen group **714** in accordance with the second aptitude level to produce the second aptitude level entigen group. For example, the answer resolution module **306** interprets the entigen information **716** to further select entigens **14**, **20**, and when including the second aptitude level to combine with entigens **2**, **5**, **8**, and **9** to form the second aptitude level entigen group **718**.

[0229] Having produced the second aptitude level entigen group **718**, the answer resolution module **306** generates a second aptitude level phrase based on the second aptitude level entigen group **718**. The second aptitude level entigen group represents a most likely interpretation of the second aptitude level phrase. The second aptitude level phrase is associated with the second aptitude level.

[0230] The generating of the second aptitude level phrase includes selecting, for each entigen of the second aptitude level entigen group, a word (e.g., any language, from the knowledge database **700**) associated with the entigen of the second aptitude level entigen group to produce the second aptitude level phrase. For example, the answer resolution module **306** outputs the second aptitude level phrase "bats are black or brown mammals that eat fruit and insects" when the second aptitude level entigen group **718** includes entigens **2**, **5**, **8**, **9**, **14**, **20**, and **30**. As another example, the answer resolution module **306** outputs a third aptitude level phrase "bats are black or brown flying mammals that eat fruit and insects" when a corresponding third aptitude level entigen group includes entigens **2**, **5**, **8**, **9**, **14**, **20**, **30**, and **50**. In another example, the answer resolution module **306** outputs a phrase of "black bats eat fruit" when the input phrase **702** includes "bats are black or brown flying mammals that eat fruit and insects" and a desired output aptitude level is less than an input aptitude level of the phrase **702**.

[0231] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0232] FIG. **10**A is a schematic block diagram of another embodiment of a computing system that includes guidance content sources **750**, the AI server **20-1** of FIG. **1**, and the user device **12-1** of FIG. **1**. The guidance content sources **750** includes the content sources **16-1** through **16-**N of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the IEI module **122** of FIG. **4**A. Generally, an embodiment of this invention presents solutions where the computing system functions to formalize content.

[0233] In an example of operation of the formalizing the content the IEI module **122** receives content for formalization. For example, the user device **12-1** issues a query request **136** to the IEI module **122**, where the query request **136** includes received text from a speech to text converter.

[0234] Having received the content for formalization, the IEI module **122** IEI processes the content to produce input knowledge. For example, for each word of a phrase of the content for formalization, the IEI module **122** identifies a set of identigens, applies identigen sequencing rules to eliminate undesired permutations of identigens, and selects a corresponding entigen to produce an entigen group that represents the input knowledge.

[0235] Having produced the input knowledge, the IEI module **122** obtains expression guidance for the content based on the input knowledge and a knowledge database that contains one or more formal rules for sentence structure and punctuation and inferred rules based on observation of textual expressions of knowledge. For example, the IEI module **122** identifies that the content includes a question in obtains expression guidance related to questions pertaining to general knowledge of connected meaning to the input knowledge (e.g., stored in the knowledge database as fact base information **600** or gathered by issuing guidance content request **752** to the kinds content sources **750**, and receiving guidance content responses **754** four further IEI processing to produce the expression guidance).

[0236] Having obtained the expression guidance, the IEI module **122** generates formalize content utilizing the expression guidance and based on one or more of the content of the input knowledge. For example, the IEI module **22** rewrites the text to include a question mark at an appropriate location of the rewritten text. Having generated the formalize content, the IEI module **122** issues a query response **140** to the user device **12-1** that includes the formalize content.

[0237] FIG. **10**B is a data flow diagram of an embodiment of formalizing content within a computing system. Input content **760** is IEI processed to produce input knowledge **762**. For example, an entigen group that includes the object entigens bats and two or more color characteristic entigen placeholders are generated when the input content **760** includes a series of words "what colors are bats".

[0238] The input knowledge **762** is utilized to access a knowledge database to obtain expression guidance knowledge **764** where the expression guidance knowledge **764** includes one or more expression entigen groups that compare favorably to the entigen group of the input knowledge **762**. The expression entigen groups include objects and characteristics associated with the entigen group of the input knowledge **762**. For example, a first entigen group that identifies that dolphins have color characteristics of black, gray, brown, and white is located as a first expression group identifying colors of an object. As another example, a second entigen group that identifies bats with color characteristics and other clarifying characteristic entigens is located as a second expression group. In particular, the second entigen group indicates that bats are black and brown and are mammals that can fly.

[0239] The input knowledge **762** and the expression guidance knowledge **764** are utilized to produce formalize content **766**, where the formalize content **766** includes punctuation and or additional clarifying and even leading facts associated with the input content **760**. As a first example, the formalized content **766** includes a sentence "What colors are bats?" to resolve the missing punctuation of the input content **760** (e.g., to include the question mark). As a second example, the formalized content **766** includes additional clarifying information from the expression group **2** in the form of a sentence "What colors are flying bats?" This distinguishes the question from a question about baseball bats (e.g., not expressly shown but anticipated to be part of the expression guidance knowledge **764**).

[0240] As a third example, the formalize content **766** includes additional leading information from the expression group **2** in the form of another sentence "Is it correct that flying bats are brown and black?" This distinguishes the question from the question about baseball bats and provides a leading answer to include knowledge from the expression guidance knowledge **764** that bats are known to be brown and black rather than to state an open question.

[0241] FIG. **10**C is a logic diagram of an embodiment of a method for formalizing content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8**L, and also FIGS. **10**A-**10**B. The method includes step **780** where a processing module of one or more processing modules of one or more computing devices of the computing system receives input content for formalization. For example, the processing module captures a live stream. As another example, the processing module extracts input content from a request to formalize the content. As yet another example, the processing module accesses content from a content storage facility.

[0242] The method continues at step **782** where the processing module IEI processes the input content to produce input knowledge. For example, for each word of a phrase of the input content, the processing module identifies a set of identigens, applies identigen sequencing rules to eliminate undesired permutations of identigens, and selects a corresponding entigen to produce an entigen group that represents the input knowledge.

[0243] The method continues at step **784** where the processing module obtains expression guidance for the input content based on the input knowledge and a knowledge database. For example, the processing module matches at least a portion of the input knowledge to a portion of the knowledge database and identifies extensions of the input knowledge based on the portion of the knowledge database and/or expression sequencing and punctuation guidance.

[0244] The method continues at step **786** where the processing module generates formalize content for the input content utilizing the expression guidance based on one or more of the input content and the input knowledge. For example, the processing module rewrites the input content and/or ads alternative knowledge for subsequent selection utilizing the expression sequencing and punctuation guidance to update the input content in accordance with the input knowledge and the portion of the knowledge database.

[0245] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0246] FIG. **11**A is a schematic block diagram of another embodiment of a computing system that includes linking content sources **800**, the AI server **20-1** of FIG. **1**, and the user device **12-1** of FIG. **1**. The linking content sources **800** includes the content sources **16-1** through **16**-N of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the IEI module **122** of FIG. **4**A. Generally, an embodiment of this invention presents solutions where the computing system functions to access a knowledge database.

[0247] In an example of operation of the accessing the knowledge database, the IEI module **122** stores knowledge generated by IEI processing content utilizing a knowledge database structure (e.g., a graphical database, a rows and columns database). Database nodes typically represent entigens that describe one of an object, a characteristic, or an action. Connectors between database nodes (e.g., entigens) represent relationships between entigens (e.g., acts on, is a, belongs to, did, did too, etc.). For example, the IEI module **122** receives linking content responses **804** from the linking content sources **800** in response to issuing one or more linking content requests **802** to the linking content sources **800**. The IEI module **122** IEI processes content of the linking content responses **804** to produce the knowledge for storage in the knowledge database structure (e.g., as fact base information **600** in the SS memory **96**).

[0248] Having stored knowledge in the knowledge database structure, the IEI module **122** generates entigen index nodes that describe the relationship between entigens of the database structure. For example, an index node links two or more similar types of entigens (e.g., link actions, link similar characteristics, link objects) of clusters of entigens or between layers of the structure. The layers of the structure is discussed in greater detail with reference to FIG. **11**B.

[0249] When accessing the knowledge database, the IEI module **122** accesses multiple levels of index nodes to facilitate searching through the database structure to find a desired cluster of entigens. For example, the IEI module **122** obtains fact base information **600** from the SS memory **96** pertaining to index nodes that link the entigens of the desired cluster of entigens.

[0250] FIG. **11**B is structure diagram of an embodiment of a knowledge database within a computing system. The knowledge database has a structure that organizes the knowledge database by levels. For example, a level 0 includes an overall anchor root node and entigen root nodes by entigen type (e.g., objects, actions, and characteristics). The anchor root node of the entire knowledge database is organized by one or more domains such that another overall root node pertains to other domain(s).

[0251] An index node level 1 includes a variety of entigen clusters associated with different aspects of an associated root node and are linked to corresponding root nodes of the index node level 0 by way of level connectors. Level connectors describe relationships between entigens by entigen type (object, characteristic, action, functional). For example, the index node level 0 root object node is linked to an entigen cluster associated with people, another entigen cluster associated with places, and another entigen cluster associated with things. As another example, the index node level 0 root characteristic node is associated with an entigen cluster associated with colors, another entigen cluster associated with sizes, and another entigen cluster associated with human expressions of emotion.

[0252] Entigen clusters of the index node level 1 are linked to entigen groups (e.g., leaf nodes) of an entigen group level. Entigen groups of the entigen group level include entigens that represent knowledge to be subsequently accessed (e.g., to generate a query response to a query etc.).

[0253] When accessing the knowledge database (e.g., to service a query) utilizing the index nodes of the structure, the database may be entered at the overall root node following a level connector to a root node associated with an entigen type based on an attribute of the query. Having traversed to the root node by entigen type, another level connector is traversed to an entigen cluster associated with that aspect of the entigen type. For example, when the query is associated with geographic locations, the traversal follows the level connector from the root object node to the entigen cluster associated with places (e.g., in the index node level 1). Further linking by connectors between entigens of the entigen cluster are utilized to follow another level connector between an entigen of the entigen cluster to an entigen group at the entigen group level to further search for knowledge associated with the query.

[0254] FIG. **11**C is a logic diagram of an embodiment of a method for accessing a knowledge database within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8**L, and also FIGS. **11**A-**11**B. The method includes step **830** where a processing module of one or more processing modules of one or more computing devices of the computing system stores knowledge in a knowledge database structure, where content is IEI processed to produce the knowledge, where the database structure includes entigens and connectors that describe relationships between the entigens. For example, the processing module obtains content, IEI processes the content to produce the knowledge, stores representations of entigens and representations of linkages between the entigens in the knowledge database (e.g., bulk knowledge stored at the entigen group level).

[0255] The method continues at step **832** where the processing module identifies commonality of the entigens of the knowledge to produce entigen index nodes for storage in the knowledge database, where the entigen index notes describe a common relationship amongst linked entigens. The identifying includes determining a type of entigen for the entigen (e.g., object, action, and characteristic), identifying a sub-type for the entigen (e.g., when an object what type of logic, when an action what type of action, when a characteristic what type of characteristic, etc.). The processing module further identifies an entigen index node of the knowledge database associated with the type and subtype, adds new entigen index nodes and/or links an existing entigen index node to entigens of the knowledge when storing the entigens of the knowledge in the knowledge database.

[0256] The method continues at step **834** where the processing module generates a query entigen group for a query. For example, the processing module IEI processes the query to produce the query entigen group. The method continues at step **836** for the processing module identifies a category of entigen index node that compares favorably to the query entigen group. For example, the processing module determines type and sub-type for an entigen of the query entigen group and matches the type and sub-type of the entigen to a corresponding entigen index node of the knowledge database.

[0257] The method continues at step **838** where the processing module accesses the knowledge database by utilizing one or more of the entigen index nodes associated with a category. For example, the processing module searches the knowledge database utilizing the corresponding entigen index node to identify a portion of the knowledge database that compares favorably to the query entigen group (e.g., to identify an answer to subsequent produce a query response).

[0258] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0259] FIG. **12**A is a schematic block diagram of another embodiment of a computing system that includes sorting content sources **850**, the AI server **20-1** of FIG. **1**, and the user device **12-1** of FIG. **1**. The sorting content sources **850** includes the content sources **16-1** through **16-**N of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the IEI module **122** of FIG. **4**A. Generally, an embodiment of this invention presents solutions where the computing system functions to access a knowledge database.

[0260] In an example of operation of the accessing the knowledge database, the IEI module **122** stores knowledge in a knowledge database structure, (e.g., as fact base information **600** in the SS memory **96**. The IEI module **122** IEI processes content to produce entigen clusters representing the knowledge. The knowledge database structure includes the entigen clusters and connectors that describe relationships between the entigens of each cluster. For example, the IEI module **122**, in response to issuing sorting content requests **854** to the sorting content sources **850**, receives sorting content responses **856**. The IEI module **122** IEI processes content of the sorting content responses **856** to produce knowledge for storage as the entigen clusters in the SS memory **96**.

[0261] The IEI module **122** links the entigen clusters via entigen index nodes associated with each linked entigen cluster. The entigen index notes describe the common relationship amongst linked entigens of the entigen clusters. The IEI module **122** generates a query entigen group for a query. For example, the IEI module **122** IEI processes query content from a query request **136** from the user device **12-1** to produce the query entigen group.

[0262] Having produced the query entigen group, the IEI module **122** identifies a category of entigen index node that compares favorably to the query entigen group. For example, the IEI module **122** obtains fact base information **600** from the SS memory **96** to identify the entigen index node.

[0263] Having identified the category, the IEI module **122** accesses the knowledge database by utilizing one or more of the entigen index nodes associated with a category. For example, the IEI module **122** obtains further fact base information **600** from the SS memory **96** to search by the category of the entigen index nodes to identify an entigen group associated with the query content. Having access the knowledge database, the IEI module **122** generates a query response **144** sending to the user device **12-1**, where the query response **140** includes one or more aspects of the accessed knowledge database pertaining to the query.

[0264] FIG. **12**B is a structure diagram of another embodiment of a knowledge database within a computing system. The knowledge database has a structure that organizes the knowledge database by levels. For example, a level 0 includes an overall anchor root node and a series of root nodes associated with any topic (e.g., to produce a very flat level 0 structure). For instance, a root truck node is established at the index node level 0 to facilitate searching for knowledge related to trucks. The anchor root node of the entire knowledge database is organized by one or more domains such that another overall root node pertains to other domain(s).

[0265] An index node level 1 includes a variety of entigen clusters associated with different aspects of an associated root node and are linked to corresponding root nodes of the index node level 0 by way of level connectors. Level connectors describe relationships between entigens by entigen type (object, characteristic, action, functional). For example, the index node level 0 root object node is linked to an entigen cluster associated with pickups, another entigen cluster associated with flatbeds, and another entigen cluster associated with box trucks.

[0266] Entigen clusters of the index node level 1 are linked to entigen groups (e.g., leaf nodes) of an entigen group level. Entigen groups of the entigen group level include entigens that represent knowledge to be subsequently accessed (e.g., to generate a query response to a query etc.). For example, pickups at index node level 1 are linked to a brand A entigen group at the entigen group level, flatbeds at index node level 1 are linked to the brand A entigen group at the entigen group level and to a brand B entigen group at the entigen group level, etc.

[0267] When accessing the knowledge database (e.g., to service a query) utilizing the index nodes of the structure, the database may be entered at the overall root node following a level connector to a root node associated with a topic associated with the query. Another connector is traversed to an entigen cluster associated with an aspect of the topic. For example, when the query is associated with trucks, the traversal follows the level connector from the root truck node to the entigen cluster associated with flatbeds (e.g., in the index node level 1). Further linking by connectors between entigens of the entigen cluster are utilized to follow another level connector between an entigen of the entigen cluster for flatbeds to an entigen group at the entigen group level to further search for knowledge associated with the query.

[0268] FIG. **12**C is a logic diagram of another embodiment of a method for accessing a knowledge database within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8**L, and also FIGS. **12**A-**12**B. The method includes step **900** where a processing module of one or more processing modules of one or more computing devices of the computing system stores knowledge in a knowledge database structure, where content is IEI processed to produce entigen clusters representing the knowledge, where the knowledge database structure includes the entigen clusters and connectors that describe relationships between the entigens of each cluster. For example, the processing module obtains the content, IEI processes the content to produce the knowledge, and stores representations of clusters of entigens of the knowledge and linkages between the entigens in the knowledge database (e.g., leaf node level).

[0269] The method continues at step **902** where the processing module links entigen clusters via entigen index nodes associated with each linked entigen cluster, where the entigen index notes describe the common relationship amongst linked entigens of the entigen clusters. For example, the processing module determines a sorting type, adds a new entigen index node when no existing entigen index node compares favorably to the sorting type, and adds linkages between the entigens and/or entigen clusters that compare favorably to the sorting type.

[0270] The method continues at step **904** where the processing module generates a query entigen group for a query. The processing module IEI processes the query to produce the query entigen group.

[0271] The method continues at step **906** where the processing module identifies a category of entigen index know the compares favorably to the query entigen group. For example, the processing module determines a type and sub-type for an entigen of the query entigen group, matches the type and/or sub-type of entigen to a corresponding entigen index node of the knowledge database.

[0272] The method continues at step **908** for the processing module accesses the knowledge database by utilizing one or more of the entigen index nodes associated with a category. For example, the processing module searches the knowledge database utilizing the corresponding entigen index node to identify a portion of the knowledge database that compares favorably to the query entigen group (e.g., to identify an answer to subsequent produce a query response).

[0273] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0274] FIG. **13**A is a schematic block diagram of another embodiment of a computing system includes reference content sources **930**, the AI server **20-1** of FIG. **1**, and the user device **12-1** of FIG. **1**. The reference content sources **930** includes the content sources **16-1** through **16-**N of FIG. **1**. The AI server **20-1** includes the processing module **50-1** of FIG. **2** and the SS memory **96** of FIG. **2**. The processing module **50-1** includes the IEI module **122** of FIG. **4**A. Generally, an embodiment of this invention presents solutions where the computing system functions to correcting impaired content.

[0275] In an example of operation of the correcting of the impaired content, the IEI module **122** receives impaired content that includes a series of words. The series of words is impaired by at least one of missing words and corrupted text. For example, a speech to text converter interprets speech incorrectly when the speech is mixed with noise from a noisy background environment. As another example, transmission of correct text is corrupted during the transmission to produce the corrupted text. For example, the IEI module **122** extracts the impaired content from a content perfection request **932** received from the user device **12-1**. For instance, the impaired content pertains to an order entry process (e.g., items for order).

[0276] The IEI module **122** IEI processes the impaired content to produce interim knowledge. For example, for each word of a phrase of the impaired content, the IEI module **122** identifies a set of identigens, applies identigen sequencing rules to eliminate undesired permutations of identigens, and selects a corresponding entigen to produce an entigen group that represents the interim knowledge.

[0277] The IEI module **122** accesses a portion of a knowledge database that compares favorably to the interim knowledge. The IEI module **122** identifies the portion of the knowledge database where at least some of the entigen group of the portion of the knowledge database aligns with the entigen group that represents the interim knowledge.

[0278] The knowledge database is produced by storing fact base information **600** and the SS memory **96** subsequent to IEI processing content of reference content responses **936** received by the reference content sources **930** to produce knowledge for the knowledge database. The IEI module **122** issues reference content request **934** to the reference content sources **932** obtained content associated with references to bolster knowledge when processing content for perfection.

[0279] Having accessed the portion of the knowledge database, the IEI module **122** generates improved knowledge based on the portion of the knowledge database. The improved knowledge adds to and/or corrects knowledge associated with the impairment of the impaired content. For example, the improved knowledge represents a viable order entry for the order entry system when the impaired content relates to an order for the order entry system.

[0280] Having generated the improved knowledge, the IEI module **122** outputs the improved knowledge. For example, the outputting includes generating a content perfection response **938** to include the improved knowledge and/or corrected content from the improved knowledge, i.e., text that adds to the series of words and/or corrects the series of words of the impaired content. Having generated the content perfection response **938**, the IEI module **122** sends the content perfection response **938** to the user device **12-1**.

[0281] As another example of operation, the IEI module **122** interprets a speech-based autonomous vehicle speech to text destination input to accurately identify the true destination. The generating of the improved knowledge may be based on a multitude of references from the reference content sources **930**, historical successful interpretations, language utilization patterns, and from reference content about destinations. Another example is discussed in greater detail with reference to FIG. **13**B.

[0282] FIG. **13**B is a data flow diagram of an embodiment for correcting impaired content within a computing system. Impaired content **950** is IEI processed to produce interim knowledge **952**. The interim knowledge **952** includes an interim knowledge entigen group that includes entigens and relationships between entigens associated with the impaired content. For example, the entigen group includes an object entigen bats, an action entigen eat, another object entigen fruit, and an open template for one or more characteristic entigens describing bats when the impaired content **950** includes a series of words “Bats are xxxx and eat fruit.” The xxxx represents a missing word of the impairment.

[0283] The interim knowledge **952** is compared to entigen groups of a knowledge database to identify a portion of the knowledge database **954** the compares favorably to the interim knowledge **952**. For example, an entigen group with entigens of connected meaning to bats is located. The entigen group includes characteristic entigens of black color, brown color, and flying mammals. The entigen group also includes entigens about eating fruit and insects.

[0284] The portion of the knowledge database **954** and the interim knowledge **952** are utilized to generate improved knowledge **956** to address the impairment of the impaired content **950**. For example, the missing word describes a characteristic of bats and the non-missing words indicate that bats eat fruit. As a result, the improved knowledge **956** must include all known characteristics of bats (e.g., black, brown, flying mammals).

[0285] The improved knowledge **956** is utilized to produce corrected content **958**. The corrected content **958** includes text of the impaired content **950** with additions and/or corrections. For example, the corrected content **958** includes a new sentence “Bats are black or brown mammals and eat fruit.” The corrected content **958** utilizes the knowledge from the portion of the knowledge database **954** to fill in for the missing word associated with characteristics of bats (e.g., they can be black or brown and are mammals).

[0286] FIG. **13**C is a logic diagram of an embodiment of a method for correcting impaired content within a computing system. In particular, a method is presented for use in conjunction with one or more functions and features described in conjunction with FIGS. **1-8**L, and also FIGS. **13**A-**13**B. The method includes step **970** where a processing module of one or more processing modules of one or more computing devices of the computing system receives impaired content that includes a string of words subject to one or more impairments (e.g., missing words, corrupted words). For example, the processing module extracts the string of words from a request to correct potentially impaired content.

[0287] The method continues at step **972** where the processing module IEI processes the impaired content to produce interim knowledge. For example, for each word of the string of words of the impaired content, the IEI module **122** identifies a set of identigens, applies identigen sequencing rules to eliminate undesired permutations of identigens, and selects a corresponding entigen to produce an entigen group that represents the interim knowledge. Alternatively, or in addition to, the processing module identifies one or more positions associated with a missing word based on the identigen pairing rules.

[0288] The method continues at step **974** where the processing module accesses a portion of a knowledge database that compares favorably to the interim knowledge. For example, the processing module identifies a portion of the knowledge database that includes an entigen group where the entigens compare favorably to the entigens of the interim knowledge.

[0289] The method continues at step **976** where the processing module generates improved knowledge utilizing the portion of the knowledge database. For example, the processing module augments the interim knowledge with additional knowledge from the portion of the knowledge database of connected meaning where the augmentation fills in for potentially missing knowledge of the interim knowledge. For example, filling in characteristics of bats when the impaired content is related to bats and is associated with characteristics of bats.

[0290] The method continues at step **978** the processing module outputs a representation of the improved knowledge. The representation of improved knowledge includes the improved knowledge and updated text representing corrected content. For example, the processing module modifies the impaired content with additional word corrections and/or additional words associated with a desired language in accordance with identigen sequencing rules based on the improved knowledge.

[0291] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0292] FIGS. **13**D-E are schematic block diagrams of another embodiment of a computing system illustrating another method for correcting impaired content within the computing system. The computing system includes the content ingestion module **300** of FIG. **5**E, the element identification module **302** of FIG. **5**E, the interpretation module **304** of FIG. **5**E, the answer resolution module **306** of FIG. **5**E, and the knowledge database **700** of FIG. **9**A. The knowledge database **700** may be implemented utilizing one or more of the memories of FIG. **2**.

[0293] FIG. **13**D illustrates an example of operation of steps of a method for correcting impaired content where a first step includes the content ingestion module **300** obtaining the impaired content **1000** for a topic that includes a plurality of words. The impairment of the impaired content **1000** is associated with one or more of a missing word, an incorrect word, an unknown word, a language mismatch word, a misspelled word, and an incorrectly ordered word. For example, the content ingestion module **300** receives the phrase that includes a first impaired word (e.g., unreadable), a second word of bats, a third word of eat, and a fourth word of fruit. In an embodiment, the content ingestion module **300** establishes an impaired content indicator when detecting the first impaired word.

[0294] Having obtained the impaired content, a second step of the example method of operation includes the content ingestion module **300** partitioning the words (e.g., utilizing a dictionary) to produce words **1002**. For example, the content ingestion module **300** partitions the impaired content **1000** to produce words **1002** that includes the impaired content indicator as a placeholder for of the first word, and the words “bats”, “eat”, and “fruit”.

[0295] Having produced the words **1002**, a third step of the example method of operation includes the element identification module **302** determining a set of identigens for each word of the plurality of words to produce a plurality of sets of identigens **708**. A set of identigens of the plurality of sets of identigens represents one or more different meanings of a word of the plurality of words. Each identigen of the set of identigens includes a meaning identifier, an instance identifier, and a time reference. Each meaning identifier associated with the set of identigens represents a different meaning of the one or more different meanings of the word of the plurality of words. Each time reference provides time information when a corresponding different meaning of the one or more different meanings is valid. A first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words.

[0296] As an example of the determining of the sets of identigens **708**, the element identification module **302** accesses, for each word of the words **1002**, the knowledge database **700** to retrieve an identigen set. For instance, the element identification module **302** receives identigen information **706** from the knowledge database **700** that includes an identigen set #**1** for the word "bats", where the identigen set #**1** includes identigens **4**, **5**, and **6**. The identigens **4**, **5**, and **6** represent three unique interpretations of the word bats, including "baseball bat", "bat (mammal)", and "to hit."

[0297] Having produced the plurality of sets of identigens **708**, a fourth step of the example method of operation includes the interpretation module **304** interpreting, in accordance with identigen pairing rules **710** of the knowledge database **700**, the plurality of sets of identigens **708** to determine a most likely meaning interpretation of the impaired content and produce an interim entigen group **1006** comprising a plurality of interim entigens. The knowledge database includes a multitude of entigen groups associated with a multitude of topics. The multitude of topics includes the topic. Each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens.

[0298] The interim entigen group represents the most likely meaning interpretation of the impaired content **1000**. Each interim entigen of the interim entigen group **1006** corresponds to a selected identigen of the set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words. Each interim entigen of the interim entigen group represents a single conceivable and perceivable thing in space and time that is independent of language and corresponds to a time reference of the selected identigen associated with the interim entigen group. The selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules **710** of the knowledge database **700**.

[0299] As an example of the producing of the interim entigen group **1006**, the interpretation module **304** identifies allowed pairings of identigens to include identigens **5** and **8**, and **8** and **9** to produce the interim entigen group **1006** to include entigens **5**, **8**, and **9**. Alternatively, or in addition to, the interpretation module **304** identifies disallowed pairings of identigens (e.g., **4** and **7**, **7** and etc.) to produce the first interim entigen group **1006** that includes entigens **5**, **8**, and **9**.

[0300] FIG. **13**E further illustrates the example method of operation of the steps for correcting impaired content, where, having produce the interim entigen group **1006**, a fifth step includes the answer resolution module **306** recovering an improved entigen group **1008** for the topic from the knowledge database **700** utilizing the interim entigen group **1006** and based on a knowledge defect of the interim entigen group caused by the impairment of the impaired content. The improved entigen group **1008** includes a plurality of improved entigens and one or more entigen relationships between at least some of the plurality of improved entigens. At least one improved entigen of the plurality of improved entigens matches a corresponding interim entigen of the plurality of interim entigens. The improved entigen group **1008** provides a beneficial cure for the knowledge defect of the interim entigen group.

[0301] The fifth step includes the answer resolution module **306** determining the knowledge defect of the interim entigen group caused by the impairment of the impaired content by one or more approaches. A first approach includes the answer resolution module **306** determining that a number of interim entigens of the interim entigen group is less than a minimum number of interim entigens threshold number. For example, the answer resolution module **306** determines the knowledge defect when the interim entigen group includes three interim entigens and the minimum number of interim entigens threshold number is four.

[0302] A second approach includes the answer resolution module **306** determining that the interim entigen group **1006** does not contain an expected yet missing interim entigen of an expected category. For example, the answer resolution module **306** determines the knowledge defect when detecting that a missing interim entigen of an expected category portraying a characteristic that describes bats is missing.

[0303] A third approach includes the answer resolution module **306** determining that the interim entigen group does not contain an expected yet missing interim entigen relationship between first and second interim entigens of the interim entigen group. For example, the answer resolution module **306** determines the knowledge defect when detecting the missing relationship between the expected yet missing interim entigen and the entigen for bats.

[0304] Having determined the knowledge defect of the interim entigen group, the recovering the improved entigen group **1008** for the topic from the knowledge database utilizing the interim entigen group **1006** and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content includes two major approaches. Each major approach includes a series of sub-steps.

[0305] A first major approach includes a first sub-step that includes the answer resolution module **306** matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database. For example, the answer resolution module **306** interprets entigen information **716** from the knowledge database **700** to match entigens **5**, **8**, and **9** of the interim entigen group to the subject entigen group from the knowledge database **700** that includes entigens **5**, **8**, and **9** (e.g., matching entigens **5**, **8**, and **9**).

[0306] A second sub-step of the first major approach includes the answer resolution module **306** identifying an extra entigen of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group. The extra entigen provides a solution for at least one of the number of interim entigens of the interim entigen group is less than the minimum number of interim entigens threshold number, and the expected yet missing interim entigen of the expected category. For example, the answer resolution module **306** interprets further entigen information **716** from the knowledge database **700** to identify the entigen **2** (e.g., black) as the extra entigen is a solution for the expected yet missing interim entigen (e.g., a characteristic of bats). Alternatively, or in addition to, the answer resolution module **306** identifies entigens **14**, **20**, and **30** as further extra entigens to include in the improved entigen group **1008**.

[0307] A third sub-step of the first major approach includes the answer resolution module **306** combining the subject entigen group with the extra entigen to produce the improved entigen group **1008**. For example, the answer resolution module **306** includes entigens **14**, **20**, and **30** along with the entigens **5**, **8**, and **9** to produce the improved entigen group **1008**.

[0308] A second major approach to recover the improved entigen group **1008** includes a first sub-step that includes the answer resolution module **306** matching at least the portion of the interim entigen group to the subject entigen group of the knowledge database as previously discussed. A second sub-step includes the answer resolution module **306** identifying the extra entigen and an extra entigen relationship of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group. The extra entigen and the extra entigen relationship provide a solution for the interim entigen group does not contain the expected yet missing interim entigen relationship between the first and second interim entigens of the interim entigen group. For example, the answer resolution module **306** interprets entigen information **716** from the knowledge database **700** to identify the entigen **2** as the extra entigen and the relationship between the entigen **2** describing the entigen **5** as the expected yet missing interim entigen relationship.

[0309] Alternatively, or in addition to, the answer resolution module **306** identifies entigens **14**, **20**, and **30** as further extra entigens and relationships between those extra entigens and entigens of the subject entigen group (e.g., **5**, **8**, and **9**) to include in the improved entigen group **1008**. Having identified the extra entigens and the missing relationships, a third sub-step of the second major approach includes the answer resolution module **306** combining the subject entigen group with the extra entigen and the extra entigen relationship to produce the improved entigen group **1008**.

[0310] Having recovered the improved entigen group **1008**, a sixth step of the example method of operation includes the answer resolution module **306** outputting, via a user interface associated with the answer resolution module **306**, a representation of the improved entigen group **1008** with an indication of an unimpaired status. The representation includes at least one of the improved entigen group **1008** and a plaintext interpretation in a first language of the improved entigen group **1008**. For example, the answer resolution module **306** outputs plaintext that includes “bats are black or brown mammals that eat fruit and insects” when the improved entigen group **1008** includes the entigens **2**, **14**, **20**, **5**, **8**, **9**, and **30**. Alternatively, or in addition to, the answer resolution module **306** sends the improved entigen group **1008** as further entigen information **716** to the knowledge database **700** for storage associated with curing impaired content.

[0311] Alternatively, or in addition to, the sixth step further includes the answer resolution module **306** combining the improved entigen group **1008** with at least one other improved entigen of the plurality of improved entigens that is not already included in the improved entigen group to produce an updated entigen group. For example, the answer resolution module **306** identifies entigen **50** (e.g., flying) as the at least one other improved entigen of the plurality of improved entigens when recovering the plurality of improved entigens from the knowledge database **700** via the entigen information **716**. Having identified the at least one other improved entigen, the answer resolution module **306** combines that at least one other improved entigen with the improved entigens **2**, **14**, **20**, **5**, **8**, **9**, and **30** to produce the updated entigen group to include entigens the entigens **2**, **14**, **50**, **20**, **5**, **8**, **9**, and **30**.

[0312] In the alternative approach, when producing the updated entigen group, the answer resolution module **306** stores the updated entigen group in the knowledge database **700** for storage associated with the curing of the impaired content. Further alternatively, the answer resolution module **306** outputs a representation of the updated entigen group. For example, the answer resolution module **306** outputs plaintext, via the user interface associated with the answer resolution module **306**, as “bats are black or brown flying mammals that eat fruit and insects.” As another example, the answer resolution module **306** outputs the updated entigen group that includes entigens **2**, **14**, **50**, **20**, **5**, **8**, **9**, and **30**.

[0313] The method described above in conjunction with the processing module can alternatively be performed by other modules of the computing system **10** of FIG. **1** or by other devices. In addition, at least one memory section (e.g., a computer readable memory, a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element etc.) that stores operational instructions can, when executed by one or more processing modules of one or more computing devices (e.g., one or more servers, one or more user devices) of the computing system **10**, cause the one or more computing devices to perform any or all of the method steps described above.

[0314] It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, audio, etc. any of which may generally be referred to as ‘data’).

[0315] As may be used herein, the terms “substantially” and “approximately” provides an industry-accepted tolerance for its corresponding term and/or relativity between items. Such an industry-accepted tolerance ranges from less than one percent to fifty percent and corresponds to, but is not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, and/or thermal noise. Such relativity between items ranges from a difference of a few percent to magnitude differences. As may also be used herein, the term(s) “configured to”, “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as “coupled to”. As may even further be used herein, the term “configured to”, “operable to”, “coupled to”, or “operably coupled to” indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term “associated with”, includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0316] As may be used herein, the term “compares favorably”, indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired relationship is that signal **1** has a greater magnitude than signal **2**, a favorable comparison may be achieved when the magnitude of signal **1** is greater than that of signal **2** or when the magnitude of signal **2** is less than that of signal **1**. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

[0317] As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network). Further note that if the processing module, module, processing circuit, and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0318] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these

functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.
[0319] To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.
[0320] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.
[0321] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.
[0322] Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.
[0323] The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules. While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

Claims

1. A method for execution by a computing device, the method comprises: obtaining impaired content for a topic that includes a plurality of words, wherein impairment of the impaired content is associated with one or more of a missing word, an incorrect word, an unknown word, a language mismatch word, a misspelled word, and an incorrectly ordered word; determining a set of identigens for each word of the plurality of words to produce a plurality of sets of identigens, wherein each set of identigens of the plurality of sets of identigens represents one or more different meanings of a corresponding word of the plurality of words, wherein each identigen of each set of identigens includes a meaning identifier, and an instance identifier, wherein each meaning identifier associated with a particular set of identigens represents a different meaning of the one or more different meanings of the corresponding word of the plurality of words, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words; interpreting, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the impaired content and produce an interim entigen group comprising a plurality of interim entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the interim entigen group represents the most likely meaning interpretation of the impaired content, wherein each interim entigen of the interim entigen group corresponds to a selected identigen of a corresponding set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each interim entigen of the interim entigen group represents a single conceivable and perceivable thing in space and time that is independent of language, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and recovering an improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on a knowledge defect of the interim entigen group caused by the impairment of the impaired content, wherein the improved entigen group includes a plurality of improved entigens and one or more entigen relationships between at least some of the plurality of improved entigens, wherein at least one improved entigen of the plurality of improved entigens matches a corresponding interim entigen of the plurality of interim entigens, wherein the improved entigen group provides a beneficial cure for the knowledge defect of the interim entigen group.

2. The method of claim 1 further comprises: combining the improved entigen group with at least one other improved entigen of the plurality of improved entigens that is not already included in the improved entigen group to produce an updated entigen group; and storing the updated entigen group in the knowledge database.

3. The method of claim 1 further comprises: outputting, via a user interface of the computing device, a representation of the improved entigen group with an indication of an unimpaired status.

4. The method of claim 1 further comprises: determining the knowledge defect of the interim entigen group caused by the impairment of the impaired content by one or more of: determining that a number of interim entigens of the interim entigen group is less than a minimum number of interim entigens threshold number; determining that the interim entigen group does not contain an expected yet missing interim entigen of an expected category; and determining that the interim entigen group does not contain an expected yet missing interim entigen relationship between first and second interim entigens of the interim entigen group.

5. The method of claim 4, wherein the recovering the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content comprises: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen provides a solution for at least one of: the number of interim entigens of the interim entigen group is less than the minimum number of interim entigens threshold number, and the expected yet missing interim entigen of the expected category; and combining the subject entigen group with the extra entigen to produce the improved entigen group.

6. The method of claim 4, wherein the recovering the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content comprises: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen and an extra entigen relationship of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen and the extra entigen relationship provide a solution for the interim entigen group that does not contain the expected yet missing interim entigen relationship between the first and second interim entigens of the interim entigen group; and combining the subject entigen group with the extra entigen and the extra entigen relationship to produce the improved entigen group.

7. A computing device of a computing system, the computing device comprises: an interface; a local memory; and a processing module operably coupled to the interface and the local memory, wherein the local memory stores operational instructions that, when executed by the processing module, causes the computing device to function to: obtain impaired content for a topic that includes a plurality of words, wherein impairment of the impaired content is associated with one or more of a missing word, an incorrect word, an unknown word, a language mismatch word, a misspelled word, and an incorrectly ordered word; determine a set of identigens for each word of the plurality of words to produce a plurality of sets of identigens, wherein each set of identigens of the plurality of sets of identigens represents one or more different meanings of a corresponding word of the plurality of words, wherein each identigen of each set of identigens includes a meaning identifier, and an instance identifier, wherein each meaning identifier associated with a particular set of identigens represents a different meaning of the one or more different meanings of the corresponding word of the plurality of words, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words; interpret, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the impaired content and produce an interim entigen group comprising a plurality of interim entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the interim entigen group represents the most likely meaning interpretation of the impaired content, wherein each interim entigen of the interim entigen group corresponds to a selected identigen of a corresponding set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each interim entigen of the interim entigen group represents a single conceivable and perceivable thing in space and time that is independent of language, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and recover, via the interface, an improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on a knowledge defect of the interim entigen group caused by the impairment of the impaired content, wherein the improved entigen group includes a plurality of improved entigens and one or more entigen relationships between at least some of the plurality of improved entigens, wherein at least one improved entigen of the plurality of improved entigens matches a corresponding interim entigen of the plurality of interim entigens, wherein the improved entigen group provides a beneficial cure for the knowledge defect of the interim entigen group.

8. The computing device of claim 7, wherein the processing module further functions to: combine the improved entigen group with at least one other improved entigen of the plurality of improved entigens that is not already included in the improved entigen group to produce an updated entigen group; and store, via the interface, the updated entigen group in the knowledge database.
9. The computing device of claim 7, wherein the processing module further functions to: output, via the interface, a representation of the improved entigen group with an indication of an unimpaired status.
10. The computing device of claim 7, wherein the processing module further functions to: determine the knowledge defect of the interim entigen group caused by the impairment of the impaired content by one or more of: determining that a number of interim entigens of the interim entigen group is less than a minimum number of interim entigens threshold number; determining that the interim entigen group does not contain an expected yet missing interim entigen of an expected category; and determining that the interim entigen group does not contain an expected yet missing interim entigen relationship between first and second interim entigens of the interim entigen group.
11. The computing device of claim 10, wherein the processing module functions to recover the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content by: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen provides a solution for at least one of: the number of interim entigens of the interim entigen group is less than the minimum number of interim entigens threshold number, and the expected yet missing interim entigen of the expected category; and combining the subject entigen group with the extra entigen to produce the improved entigen group.
12. The computing device of claim 10, wherein the processing module functions to recover the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content by: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen and an extra entigen relationship of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen and the extra entigen relationship provide a solution for the interim entigen group that does not contain the expected yet missing interim entigen relationship between the first and second interim entigens of the interim entigen group; and combining the subject entigen group with the extra entigen and the extra entigen relationship to produce the improved entigen group.
13. A non-transitory computer readable memory comprises: a first memory element that stores operational instructions that, when executed by a processing module, causes the processing module to: obtain impaired content for a topic that includes a plurality of words, wherein impairment of the impaired content is associated with one or more of a missing word, an incorrect word, an unknown word, a language mismatch word, a misspelled word, and an incorrectly ordered word; a second memory element that stores operational instructions that, when executed by the processing module, causes the processing module to: determine a set of identigens for each word of the plurality of words to produce a plurality of sets of identigens, wherein each set of identigens of the plurality of sets of identigens represents one or more different meanings of a corresponding word of the plurality of words, wherein each identigen of each set of identigens includes a meaning identifier and an instance identifier, wherein each meaning identifier associated with a particular set of identigens represents a different meaning of the one or more different meanings of the corresponding word of the plurality of words, wherein a first set of identigens of the plurality of sets of identigens is produced for a first word of the plurality of words; a third memory element

that stores operational instructions that, when executed by the processing module, causes the processing module to: interpret, in accordance with identigen pairing rules of a knowledge database, the plurality of sets of identigens to determine a most likely meaning interpretation of the impaired content and produce an interim entigen group comprising a plurality of interim entigens, wherein the knowledge database includes a multitude of entigen groups associated with a multitude of topics, wherein the multitude of topics includes the topic, wherein each entigen group of the multitude of entigen groups includes a corresponding plurality of entigens and one or more entigen relationships between at least some of the corresponding plurality of entigens, wherein the interim entigen group represents the most likely meaning interpretation of the impaired content, wherein each interim entigen of the interim entigen group corresponds to a selected identigen of a corresponding set of identigens having a selected meaning of the one or more different meanings of each word of the plurality of words, wherein each interim entigen of the interim entigen group represents a single conceivable and perceivable thing in space and time that is independent of language, wherein the selected identigen favorably pairs with at least one corresponding sequentially adjacent identigen of another set of identigens of the plurality of sets of identigens based on the identigen pairing rules of the knowledge database; and a fourth memory element that stores operational instructions that, when executed by the processing module, causes the processing module to: recover an improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on a knowledge defect of the interim entigen group caused by the impairment of the impaired content, wherein the improved entigen group includes a plurality of improved entigens and one or more entigen relationships between at least some of the plurality of improved entigens, wherein at least one improved entigen of the plurality of improved entigens matches a corresponding interim entigen of the plurality of interim entigens, wherein the improved entigen group provides a beneficial cure for the knowledge defect of the interim entigen group.

14. The non-transitory computer readable memory of claim 13 further comprises: a fifth memory element that stores operational instructions that, when executed by the processing module, causes the processing module to: combine the improved entigen group with at least one other improved entigen of the plurality of improved entigens that is not already included in the improved entigen group to produce an updated entigen group; and store the updated entigen group in the knowledge database.

15. The non-transitory computer readable memory of claim 13 further comprises: a sixth memory element that stores operational instructions that, when executed by the processing module, causes the processing module to: output a representation of the improved entigen group with an indication of an unimpaired status.

16. The non-transitory computer readable memory of claim 13 further comprises: a seventh memory element that stores operational instructions that, when executed by the processing module, causes the processing module to: determine the knowledge defect of the interim entigen group caused by the impairment of the impaired content by one or more of: determining that a number of interim entigens of the interim entigen group is less than a minimum number of interim entigens threshold number; determining that the interim entigen group does not contain an expected yet missing interim entigen of an expected category; and determining that the interim entigen group does not contain an expected yet missing interim entigen relationship between first and second interim entigens of the interim entigen group.

17. The non-transitory computer readable memory of claim 16, wherein the processing module function to execute the operational instructions stored by the fourth memory element to cause the processing module to recover the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content by: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen provides a solution for at least one of: the number of interim entigens of the interim entigen group is less than the minimum number of interim entigens threshold number, and the expected yet missing interim entigen of the expected category; and combining the subject entigen group with the extra entigen to produce the improved entigen group.

18. The non-transitory computer readable memory of claim 16, wherein the processing module function to execute the operational instructions stored by the fourth memory element to cause the processing module to recover the improved entigen group for the topic from the knowledge database utilizing the interim entigen group and based on the knowledge defect of the interim entigen group caused by the impairment of the impaired content by: matching at least a portion of the interim entigen group to a subject entigen group of the knowledge database; identifying an extra entigen and an extra entigen relationship of the knowledge database that is associated with the subject entigen group yet the extra entigen is excluded from the subject entigen group, wherein the extra entigen and the extra entigen relationship provide a solution for the interim entigen group that does not contain the expected yet missing interim entigen relationship between the first and second interim entigens of the interim entigen group; and combining the subject entigen group with the extra entigen and the extra entigen relationship to produce the improved entigen group.